

Spatially Resolved Transcriptomics in Male Mouse Kidney After Ischemia-Reperfusion Injury

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Background

Inducing Ischemia:

- The renal pedicles (the bundle of vessels) are clamped with non-traumatic microaneurysm clamps.
- For male samples, the clamp time was 22 minutes

Reperfusion & Recovery:

- After the 22-minute clamp time, the clamps are removed, allowing blood flow to resume

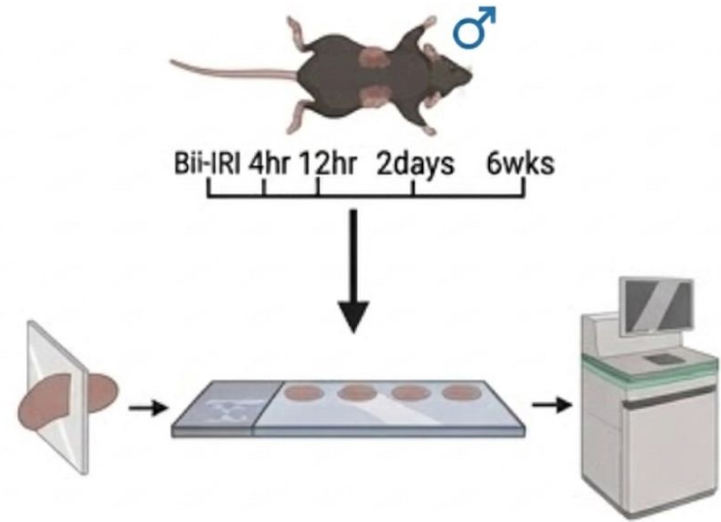
Datapoints:

- Sham: Mice undergo the same surgical procedure except the renal pedicles are not clamped.
- 4 hours post-IRI: Early acute phase. Tissue collected 4 hours after clamps were removed.
- 12 hours post-IRI: Acute injury phase. Tissue collected 12 hours after clamps were removed.

Kidney Dissection:

- Kidneys are removed, and the renal capsule is taken off.
- Each kidney is *bisected coronally*

Goal: Induce a controlled, reversible injury to the kidneys to study the transcriptional response to ischemia and subsequent repair.



De Novo Transcriptional Induction

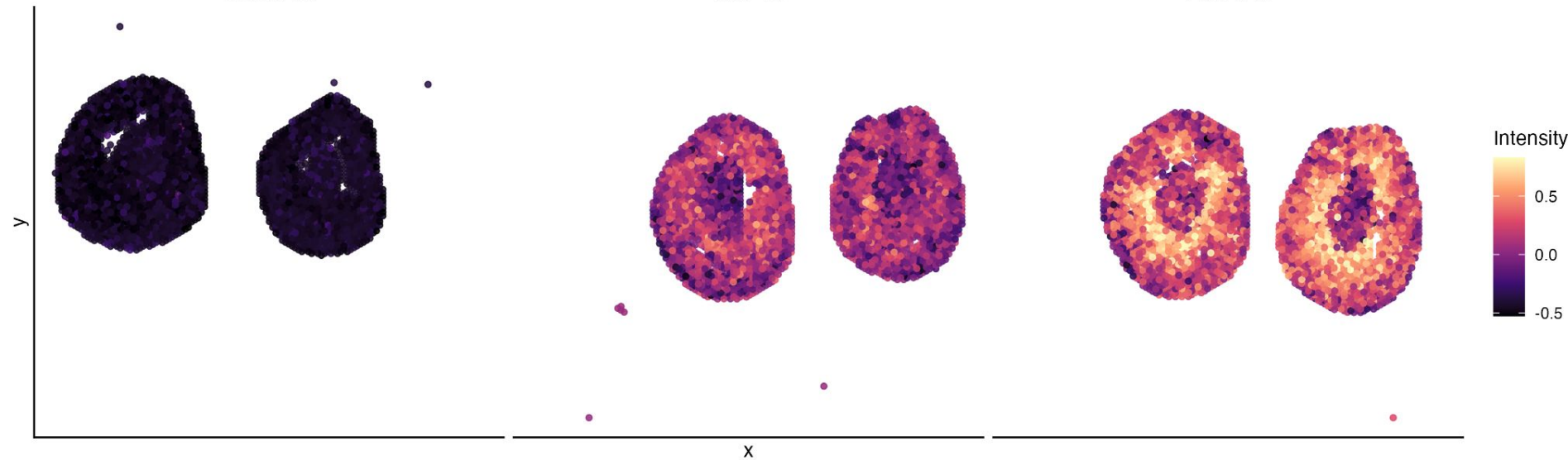
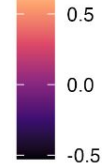
Spatial mapping of genes dormant at baseline reveals aggressive activation post-injury.

Sham 0h

AKI 4h

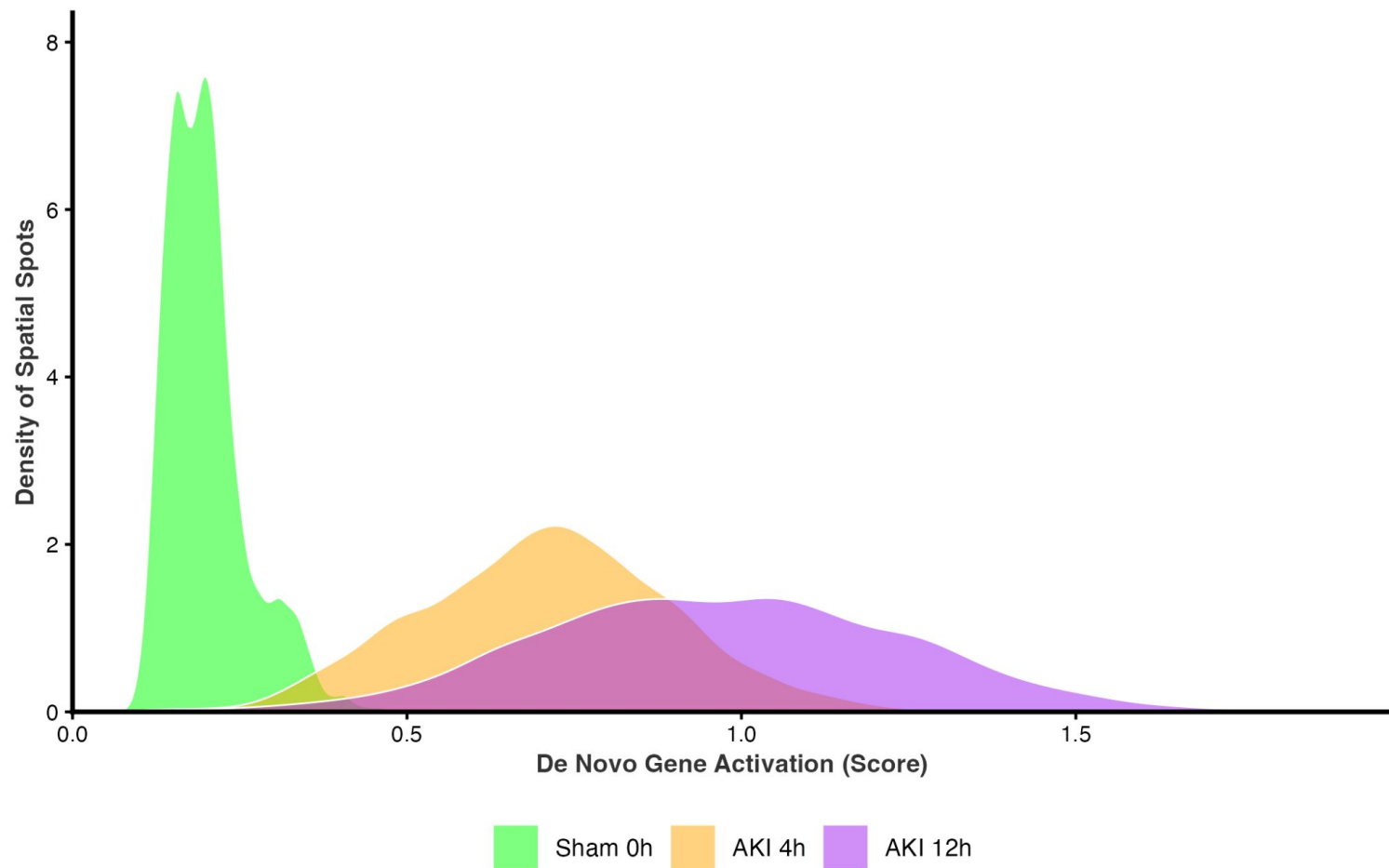
AKI 12h

Intensity



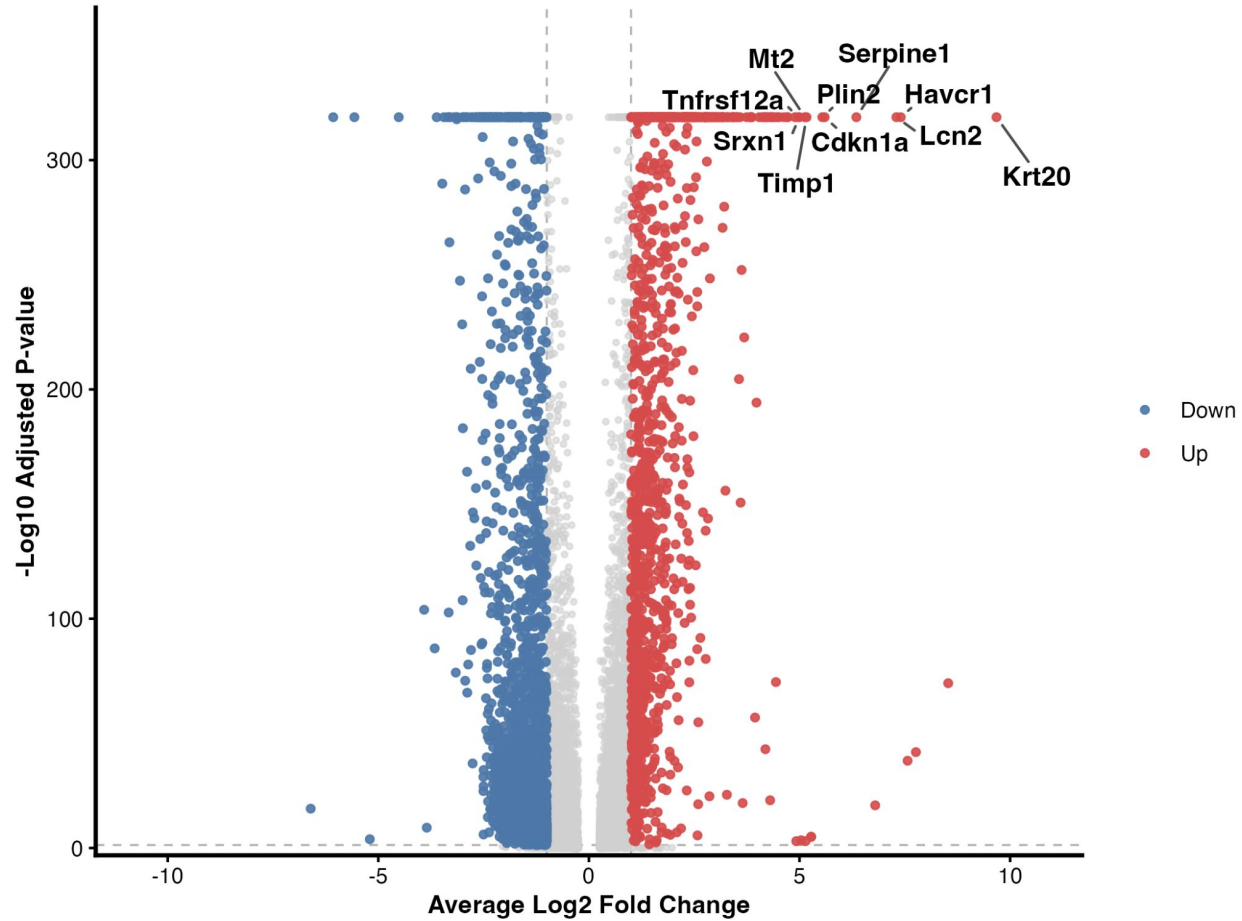
Global Induction of the Injury Transcriptome

Zero-anchored density proves the global transition from baseline silence to peak 12h eruption.



Discovery of the Injury Transcriptome

Differential expression analysis identifies top upregulated markers of acute cortical stress.



Significance: Adjusted P-value < 0.05 and $|\log_2 \text{FC}| > 1.0$

Integrated Damage Score

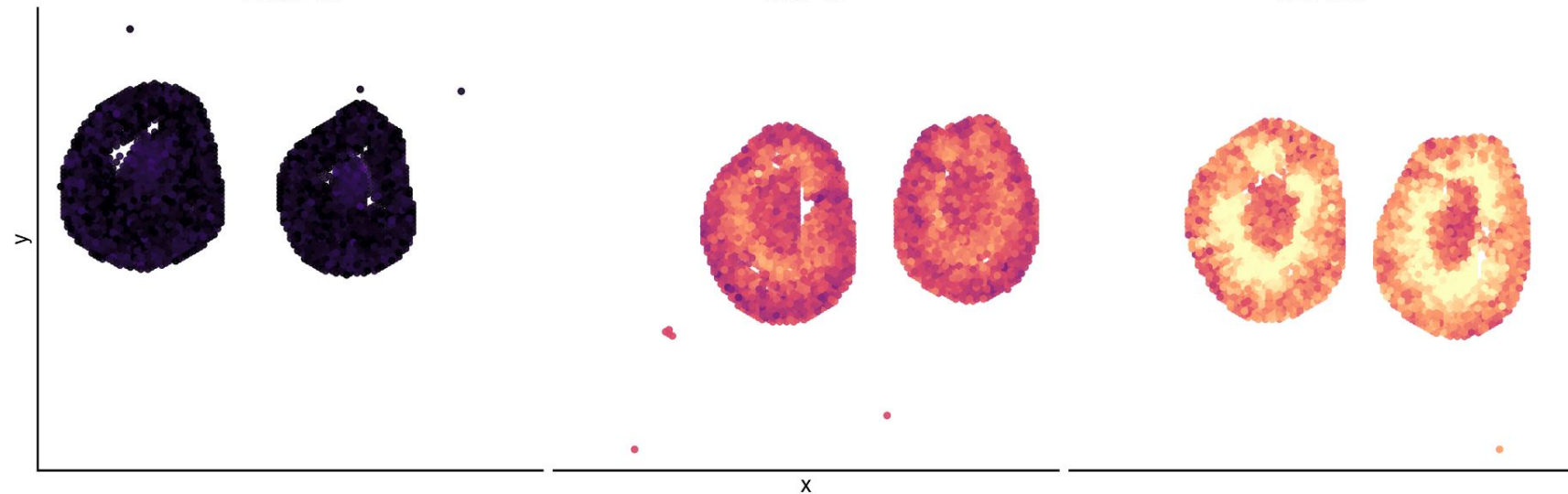
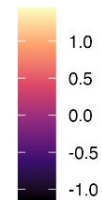
Aggregated expression of top 10 discovery markers localizes the injury epicenter.

Sham 0h

AKI 4h

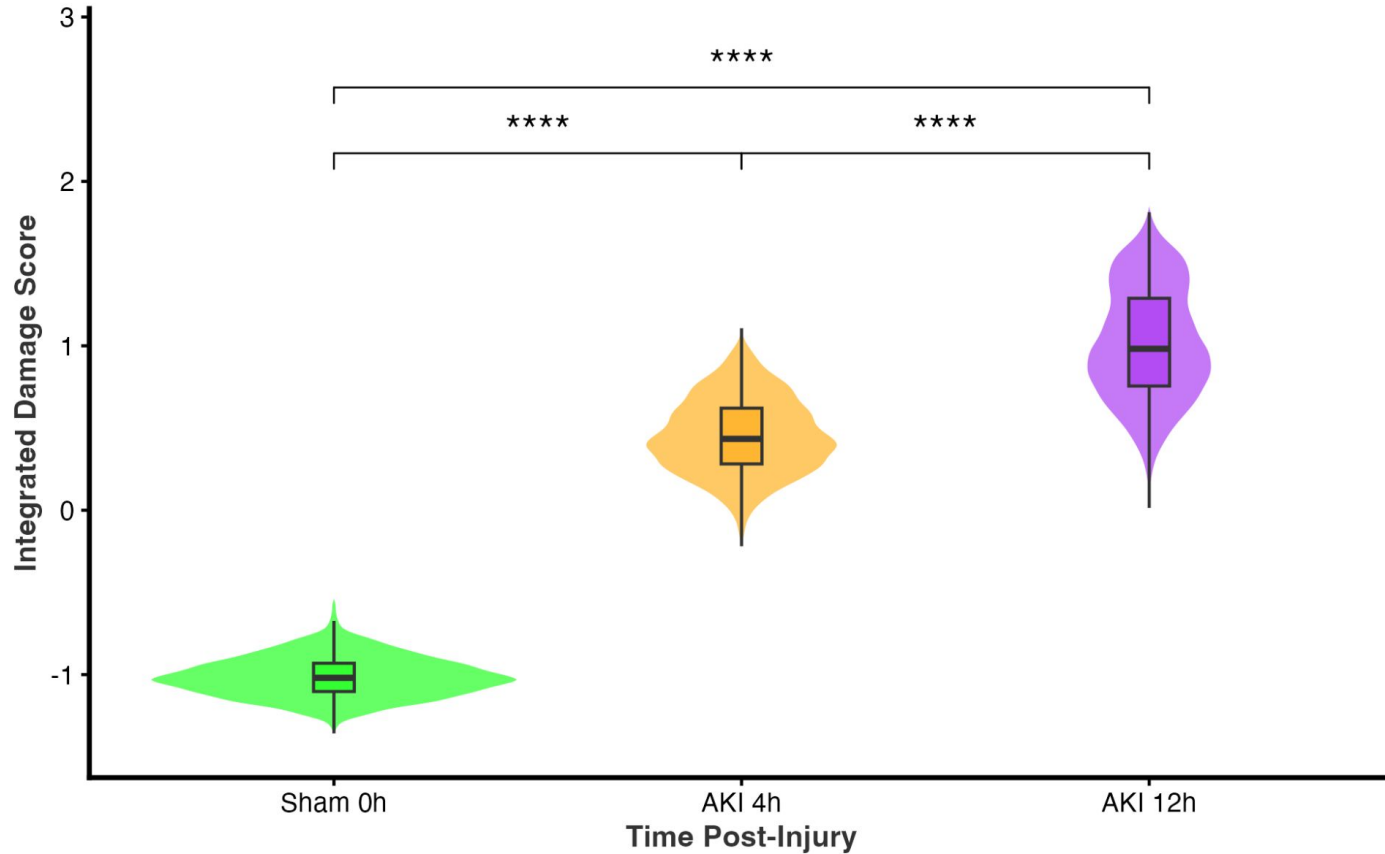
AKI 12h

Damage Score
(SCT Module)



Statistical Validation of Tissue Damage

Pairwise comparisons (t-test) confirm the significant escalation of damage over time.

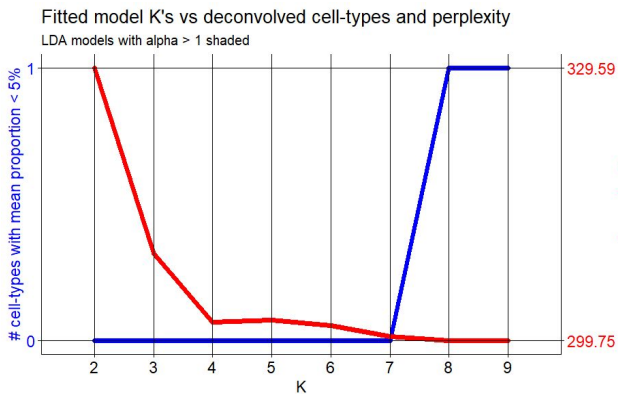


STDeconvolve

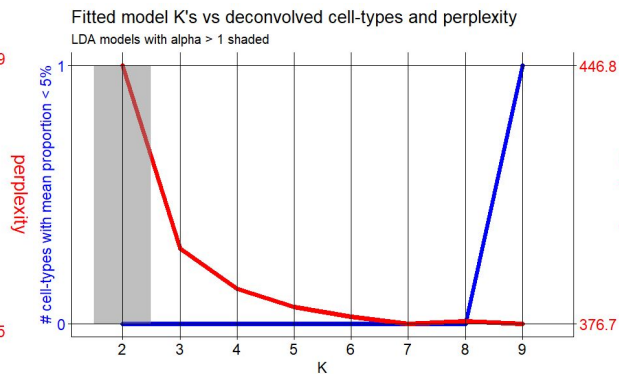


Determining STdeconvolve Cluster Numbers

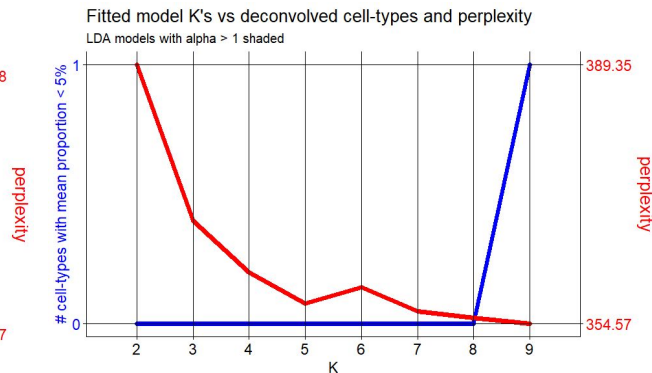
Sham



4 Hours

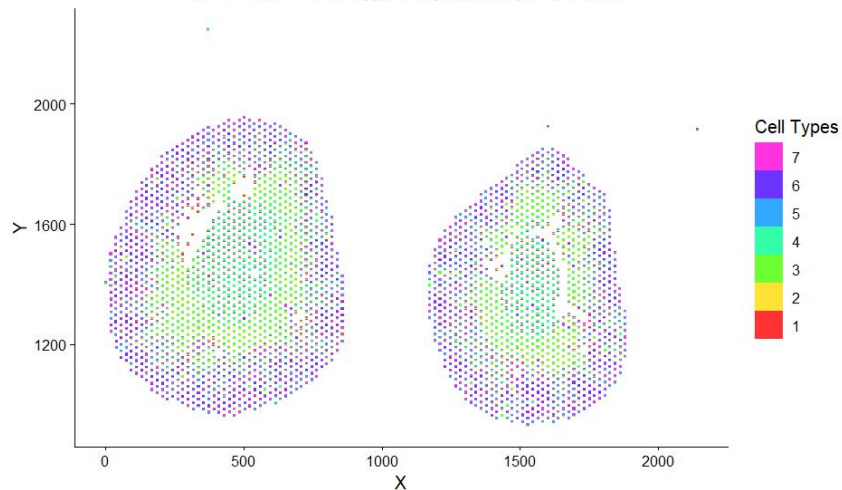


12 Hours

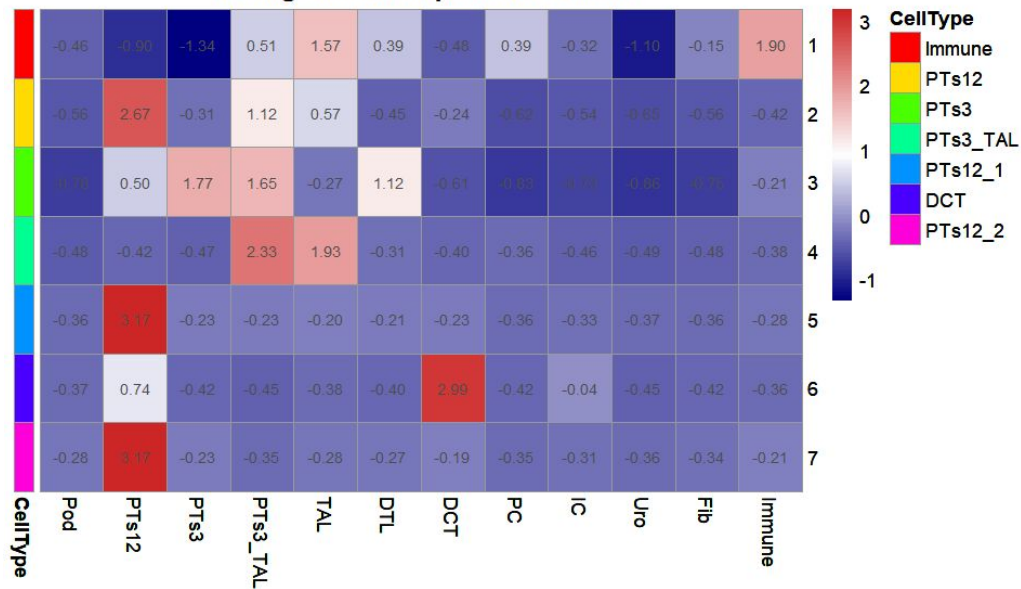


Cell Type Annotations for Sham IRI

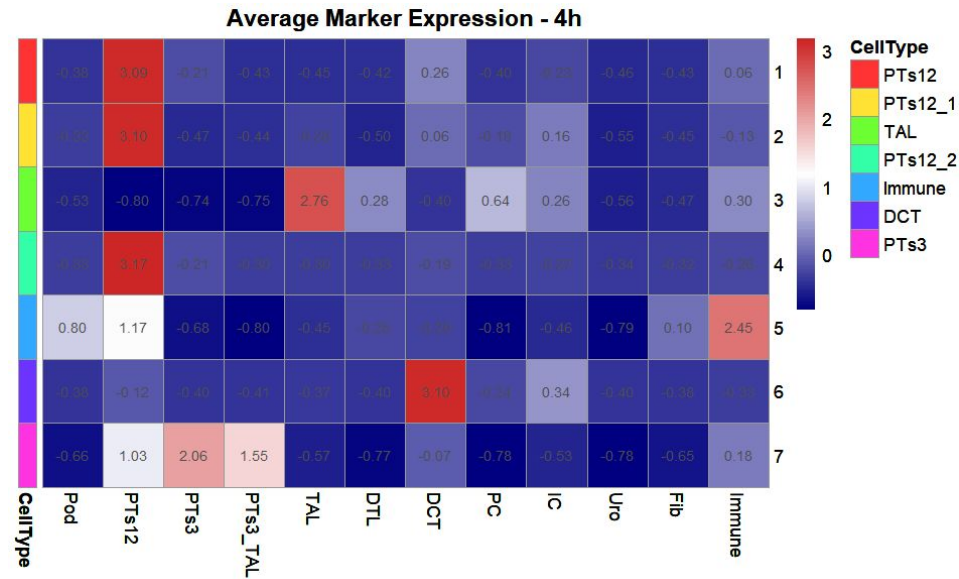
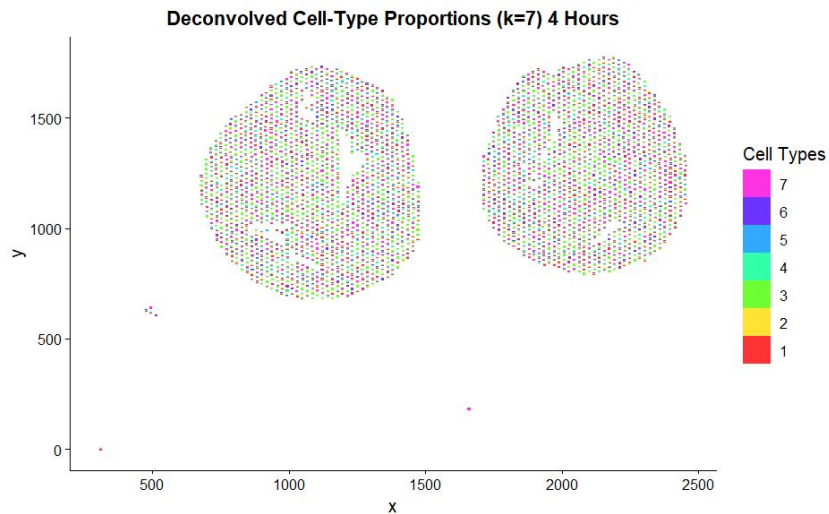
Deconvolved Cell-Type Proportions (k=7) Sham



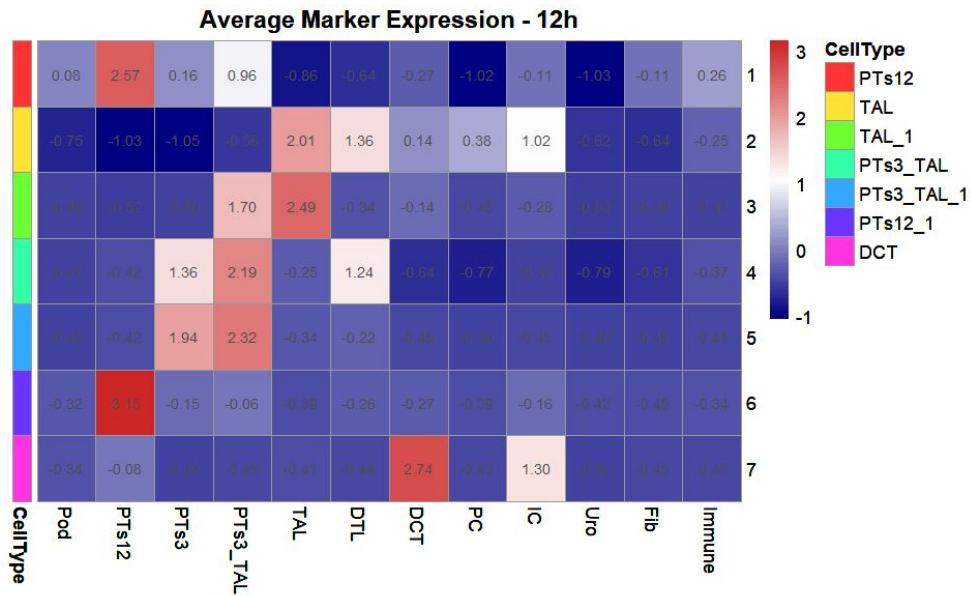
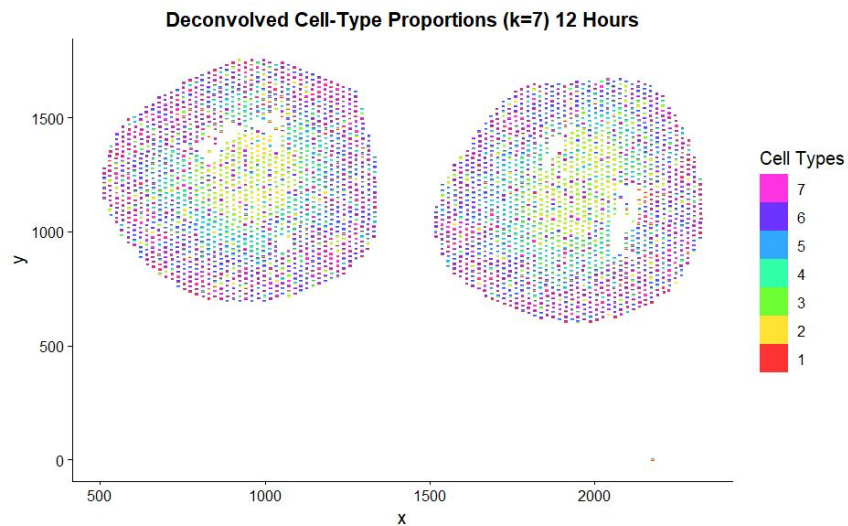
Average Marker Expression - Sham



Cell Type Annotations for 4 Hours IRI

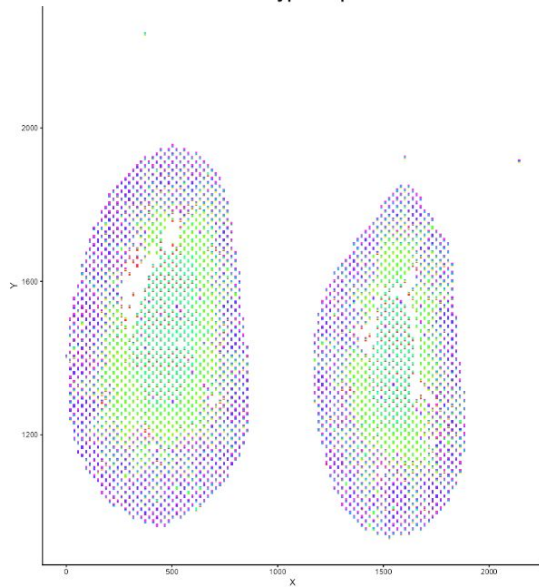


Cell Type Annotations for 12 Hours IRI

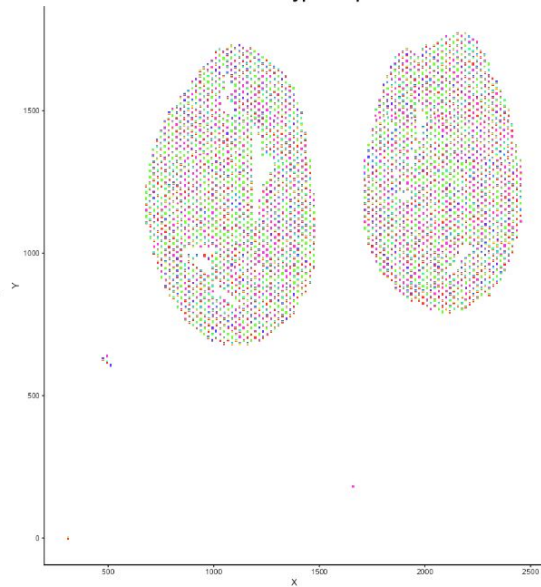


STdeconvolve Topic Annotations

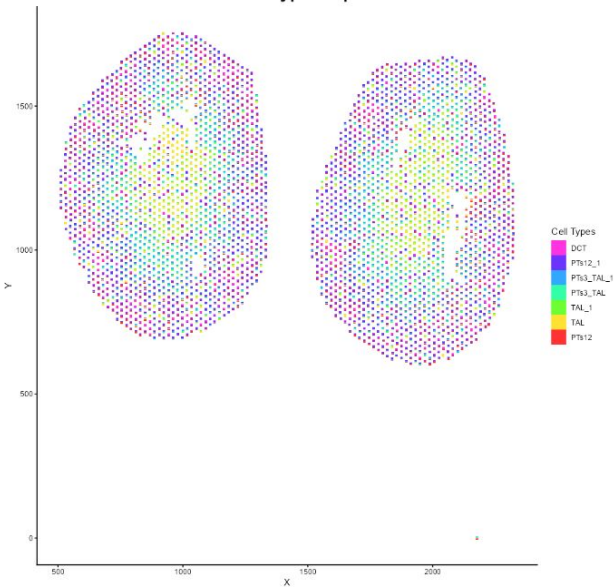
Deconvolved Cell-Type Proportions: Sham



Deconvolved Cell-Type Proportions: 4h



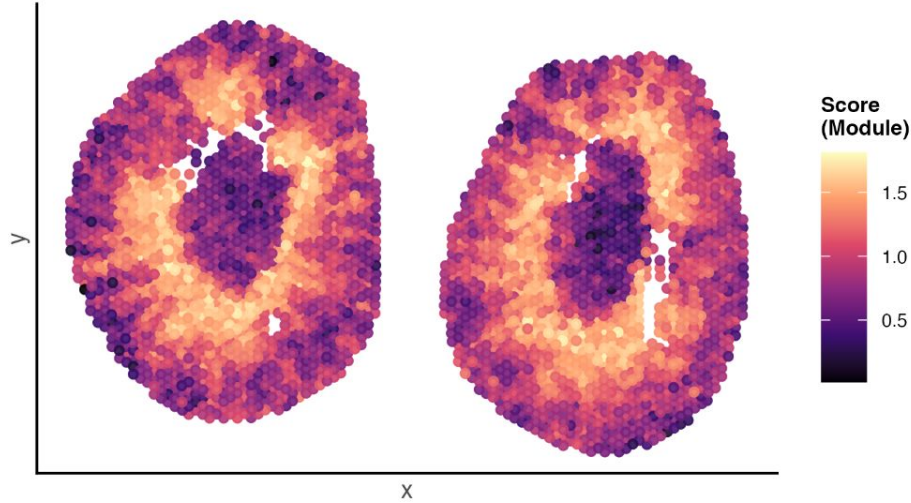
Deconvolved Cell-Type Proportions: 12h



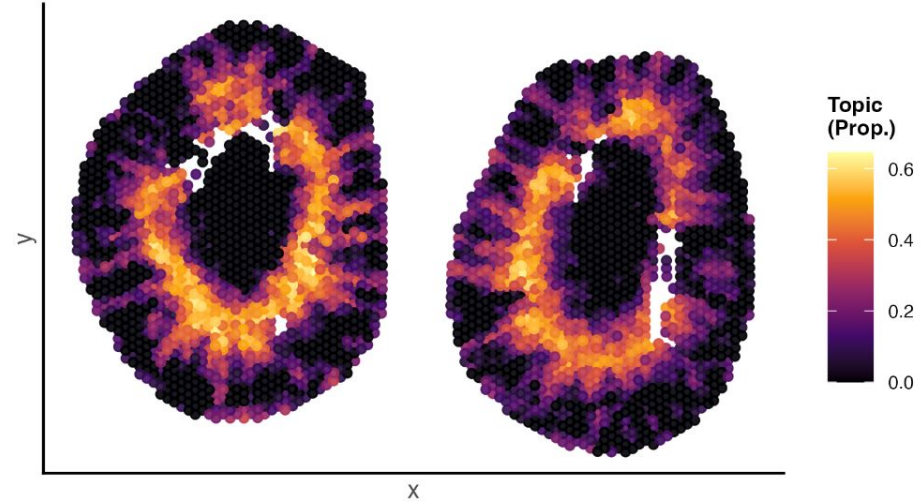
Discovery vs. Deconvolution Alignment

Spatiotemporal overlap between top damage markers and the deconvoluted injury topic.

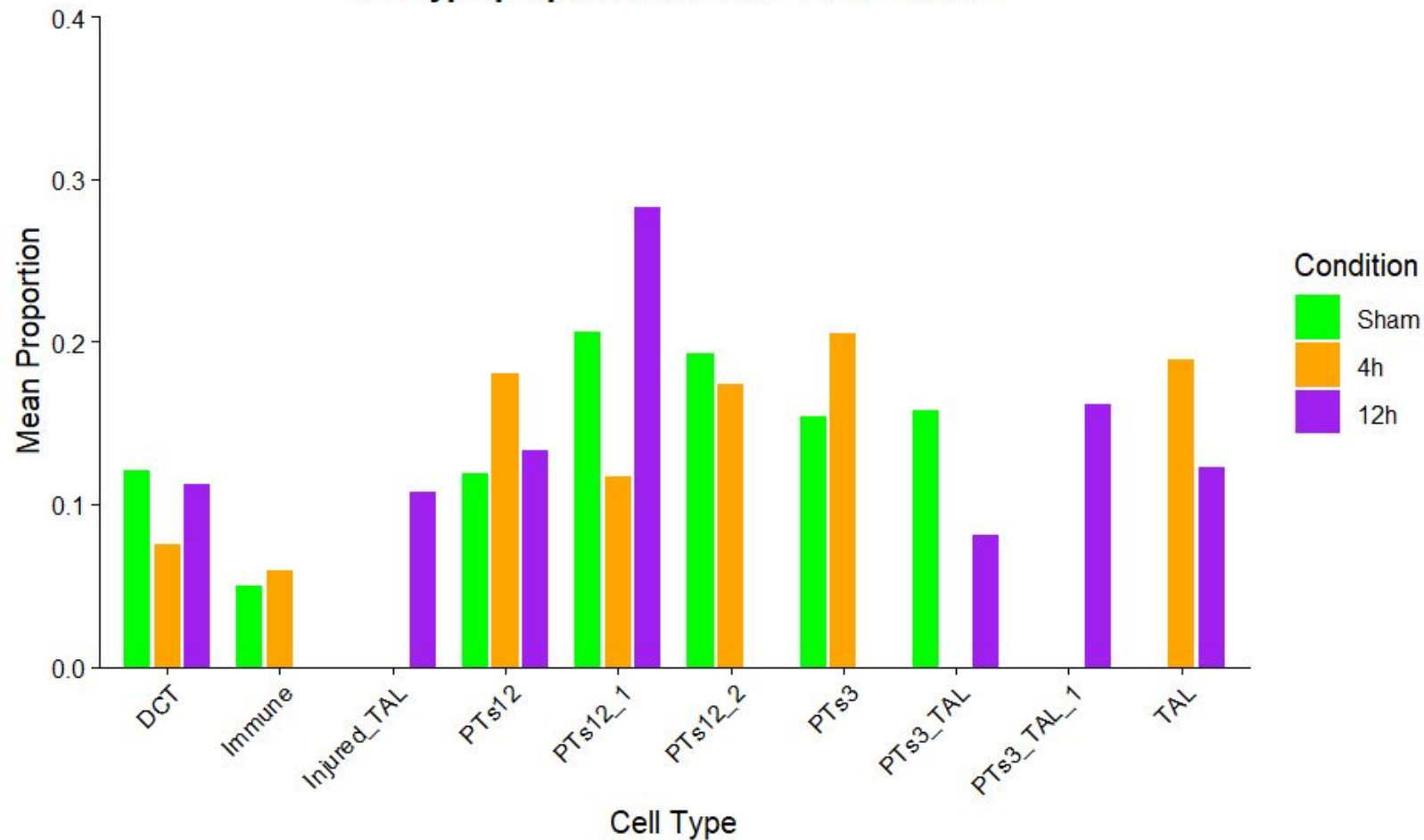
1. Damage Epicenter



2. Topic 5

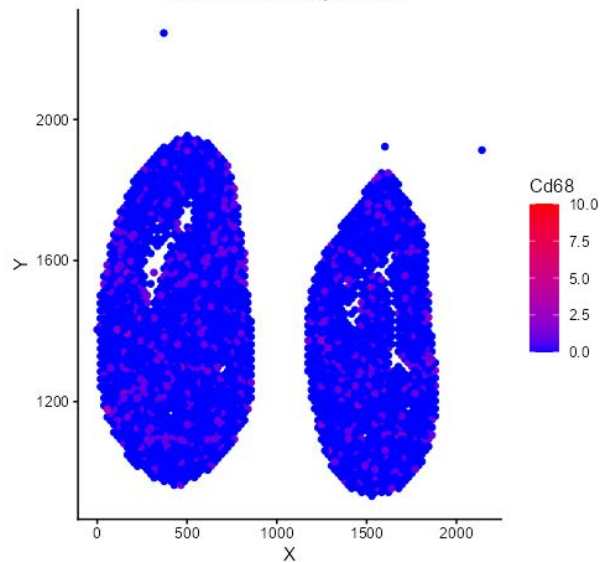


Cell type proportions: Sham vs 4h vs 12h

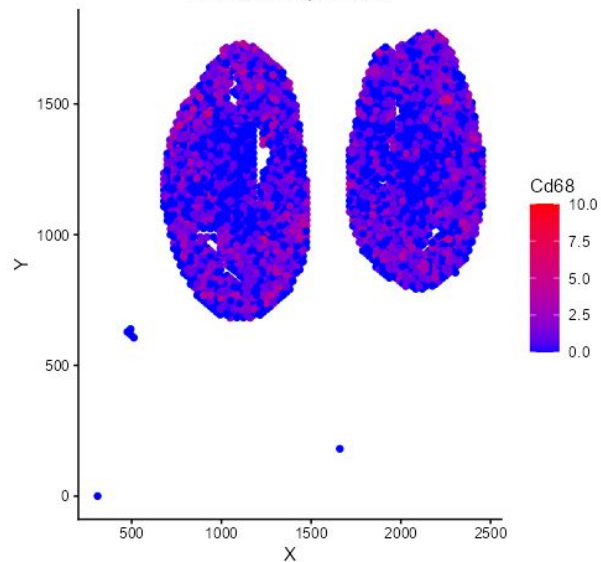


There *are* macrophages in the data!!!

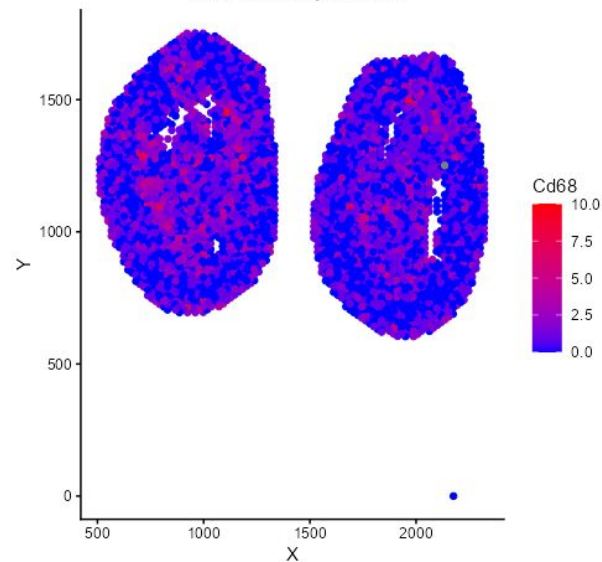
Sham: CD68 Expression



4h: CD68 Expression

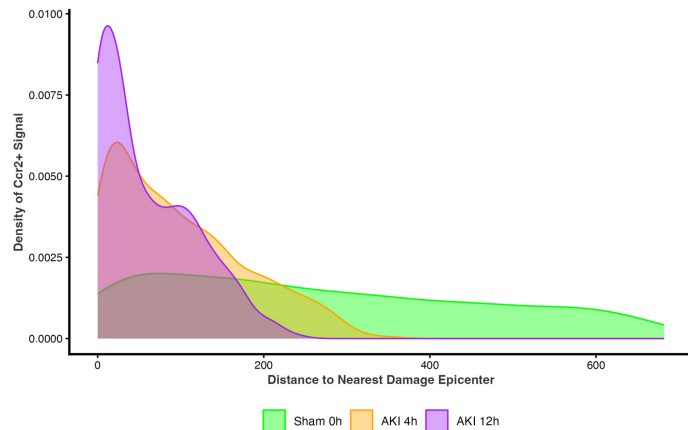


12h: CD68 Expression

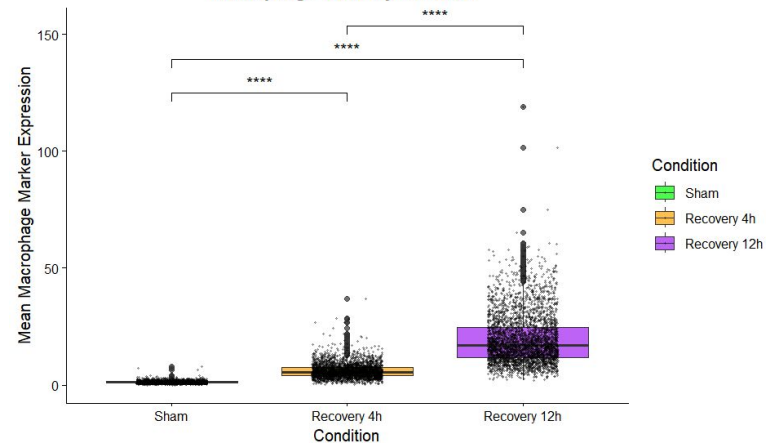


Mathematical Proof of Targeted Immune Migration

At 12h, the physical distance between the immune receptors and damage epicenters collapses to zero.

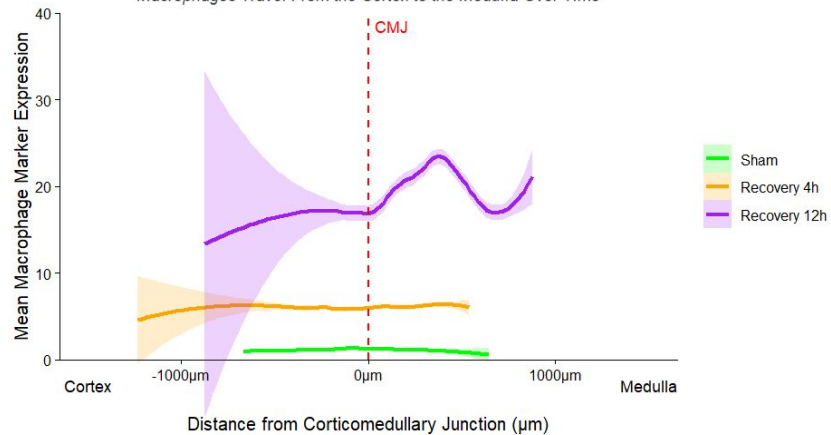


Macrophage Score by Condition

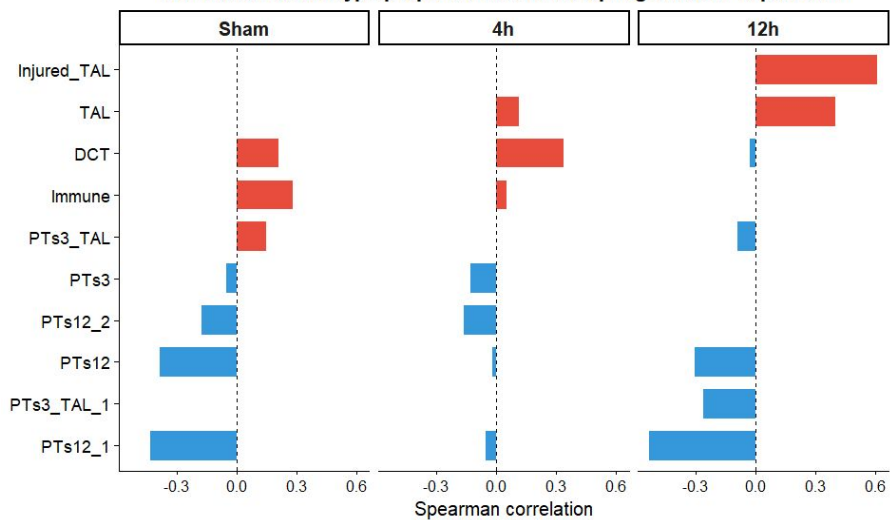


Macrophage Expression vs Distance from CMJ

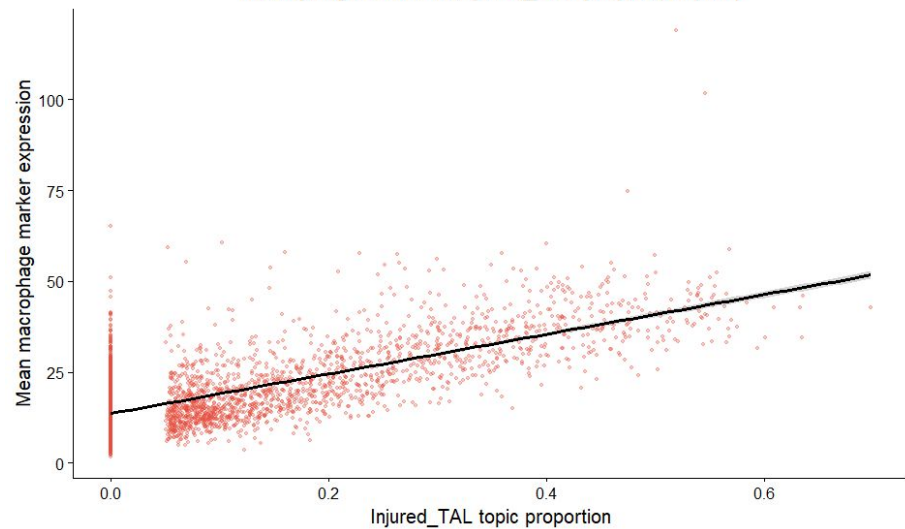
Macrophages Travel From the Cortex to the Medulla Over Time



Correlation of cell type proportion with macrophage marker expression



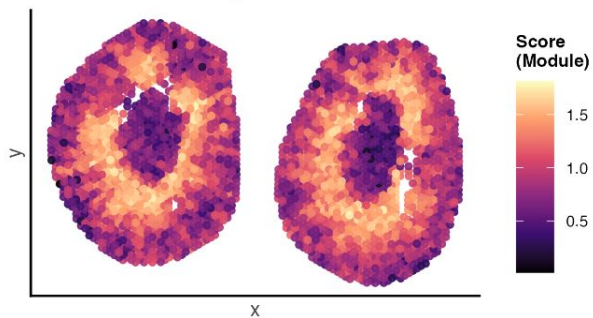
Macrophage score vs Injured_TAL proportion (12h)



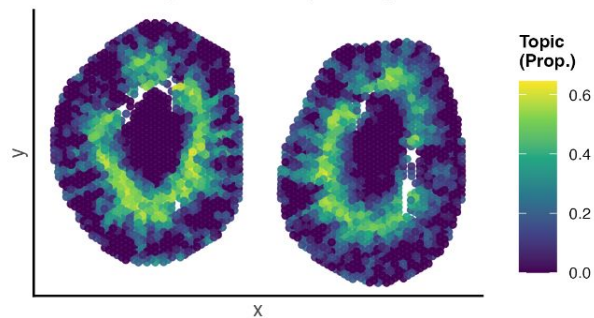
Evaluating Spatial Correlation at 12 Hours

Spatiotemporal entanglement between tissue damage, neighborhood remodeling, and immune infiltration.

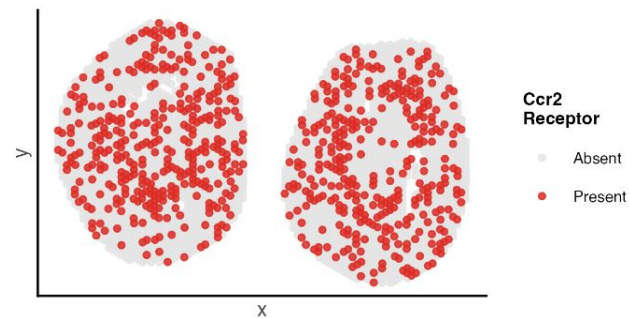
1. Damage Epicenter



2. Hybrid Niche (TAL_1)

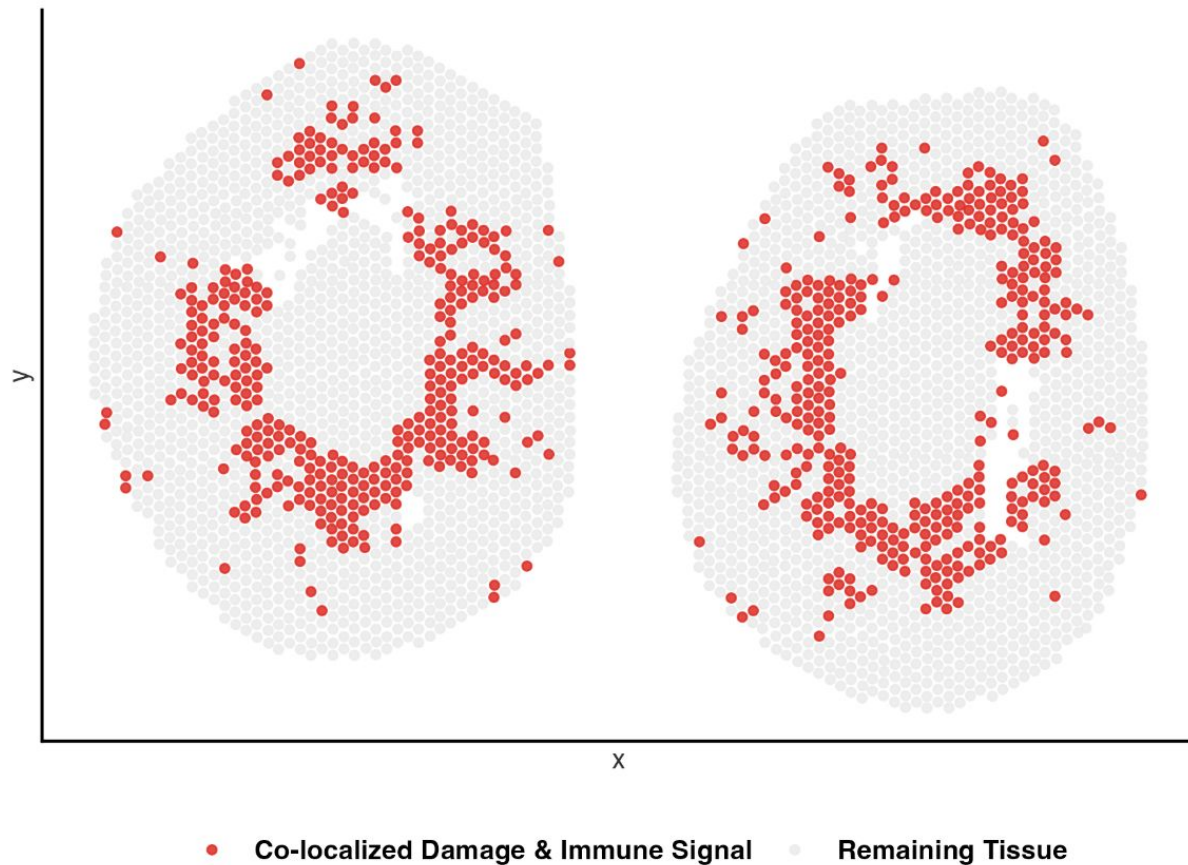


3. Immune Influx



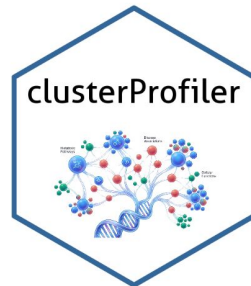
Bivariate Spatial Association (LISA Map)

Statistical identification of territories where tissue damage and immune signaling co-localize.

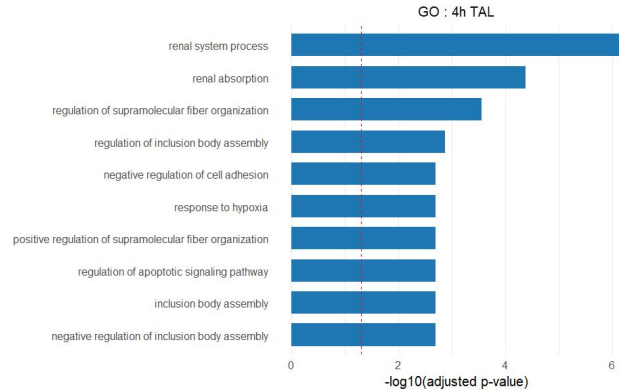
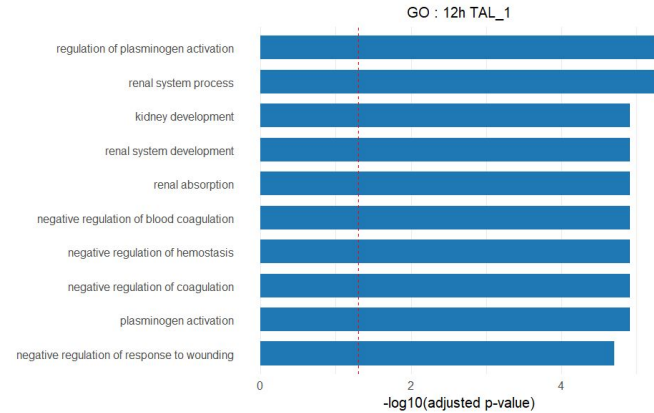
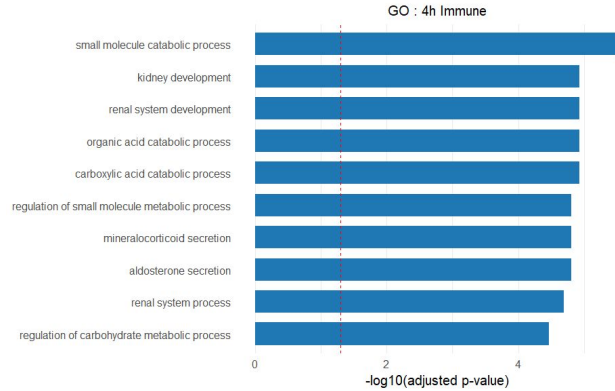
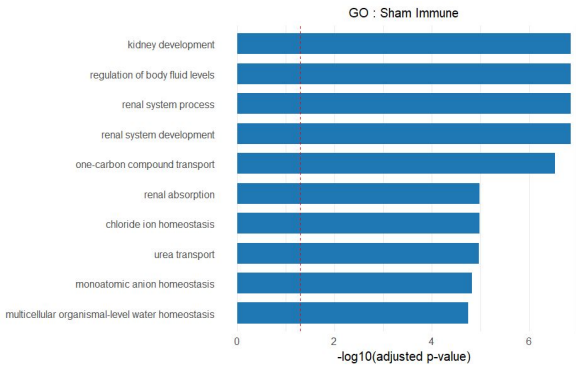


Enrichment Analysis

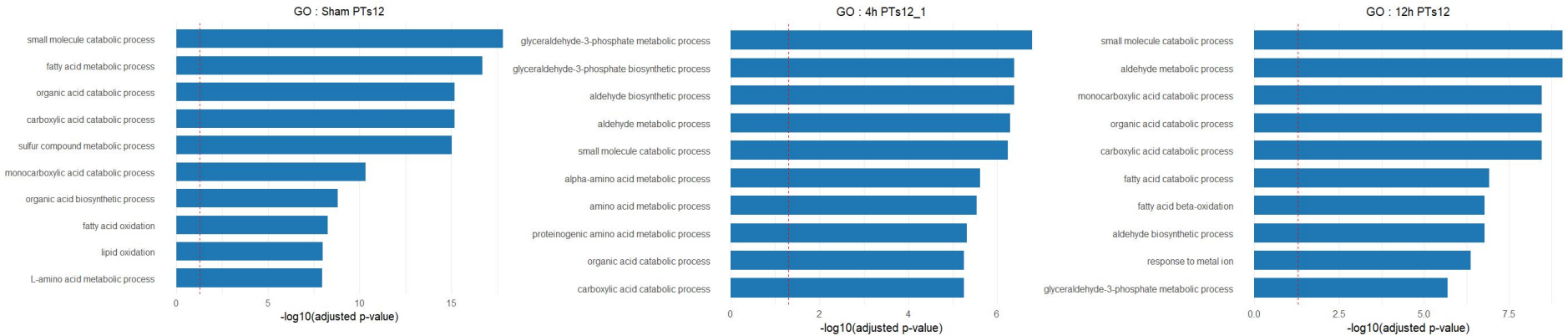
- Identify if specific **sets** of genes or **pathways** show statistically significant differences
- requires predefined gene sets
- Gene Ontology (GO) is a database with standardized terms that categorize genes and gene products based on their biological functions, processes, and cellular components
- tests whether certain GO terms appear *more frequently* in a list of genes than would be expected by chance



Enrichment Analysis on Immune Cells and Thick Ascending Limb



Enrichment Analysis on Straight Proximal Tubules



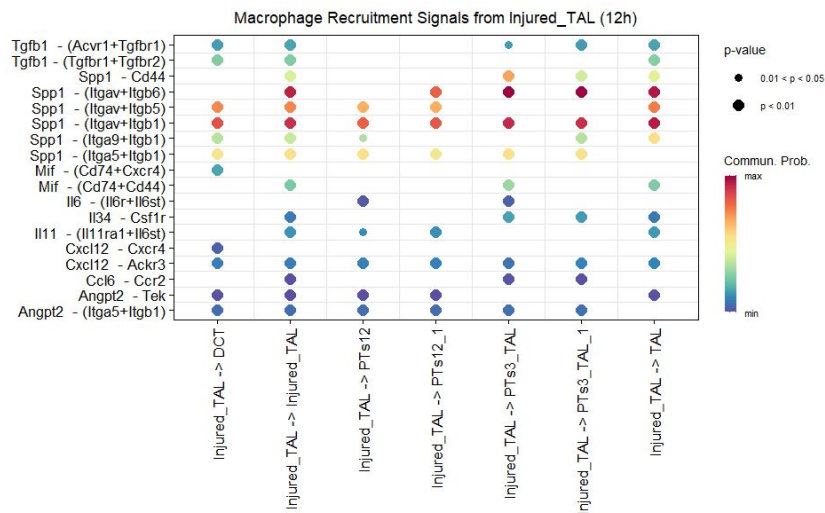
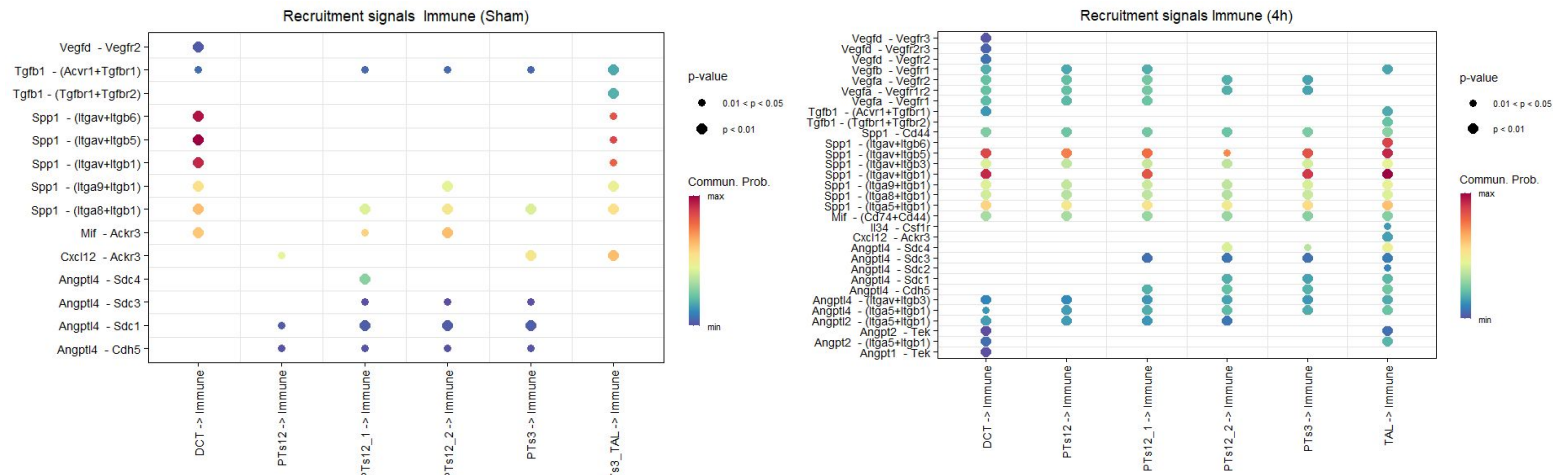
CellChat

- infers cell-cell communication by looking at expression of ligand-receptor pairs

CellChat looks at:

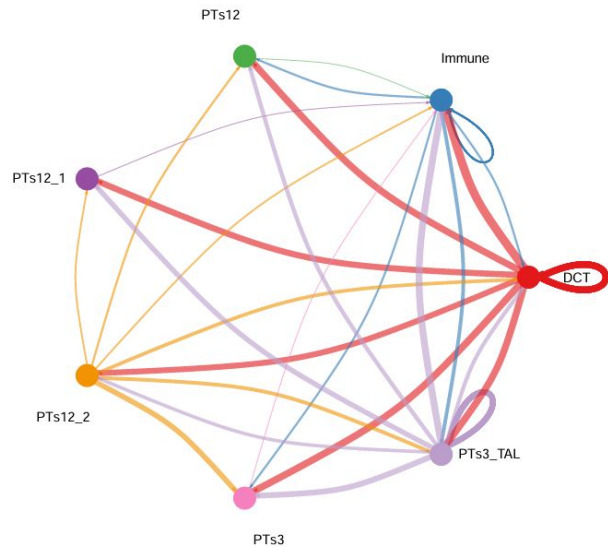
- Which cells are present (from our STdeconvolve cell type annotations)
- Pairwise analysis of each cell type to see if sender has ligands and if receiver has matching receptors



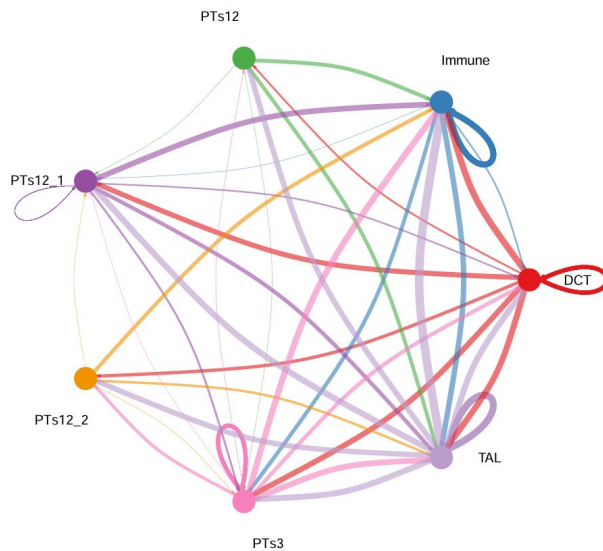


SPP1 Signaling Pathway Network (Immune Recruitment)

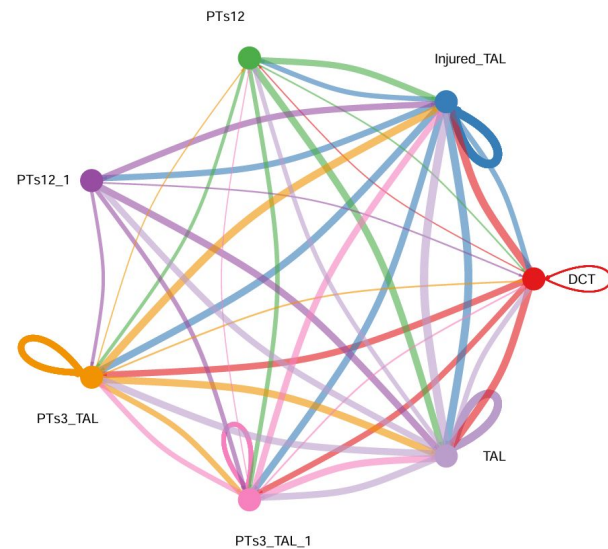
Sham



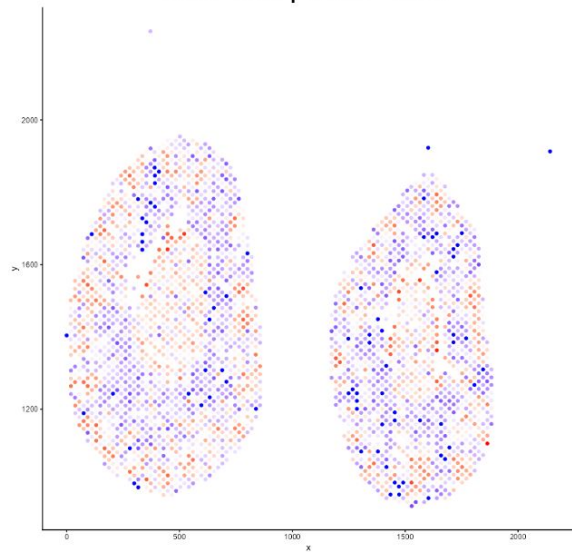
4 Hours



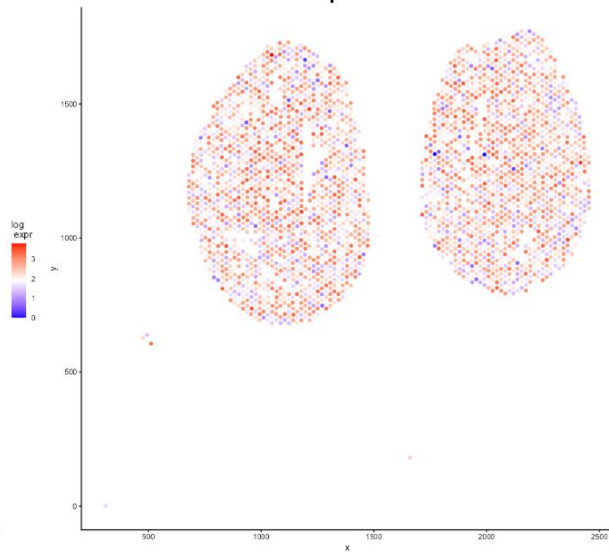
12 Hours



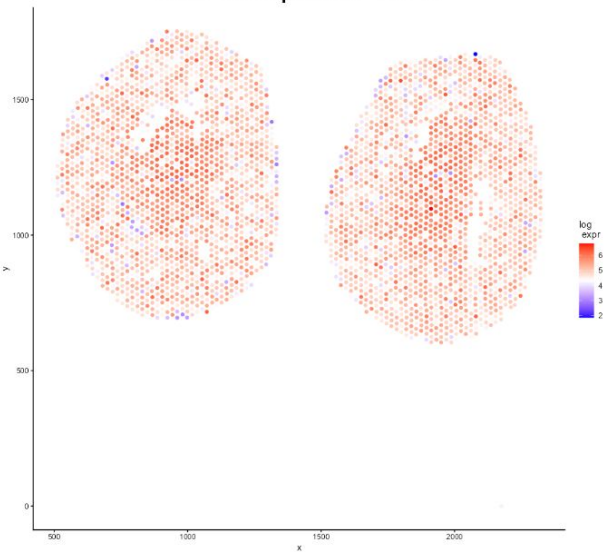
Spp1 - Sham
max=3.81 | mean=1.744



Spp1 - 4h
max=5.32 | mean=3.528

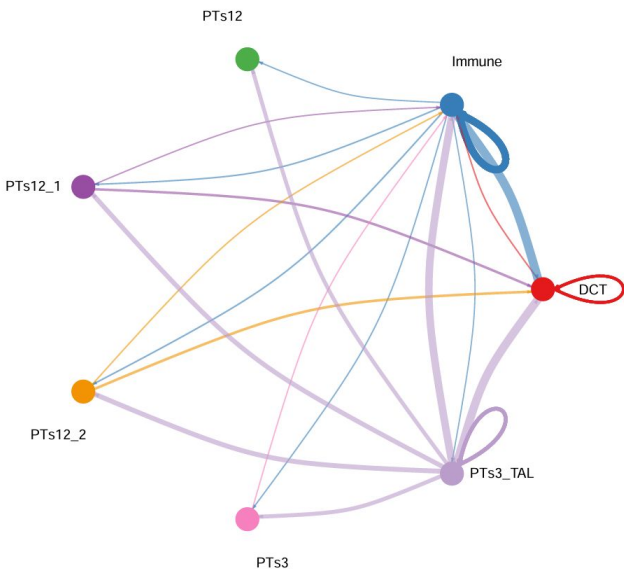


Spp1 - 12h
max=6.79 | mean=5.044

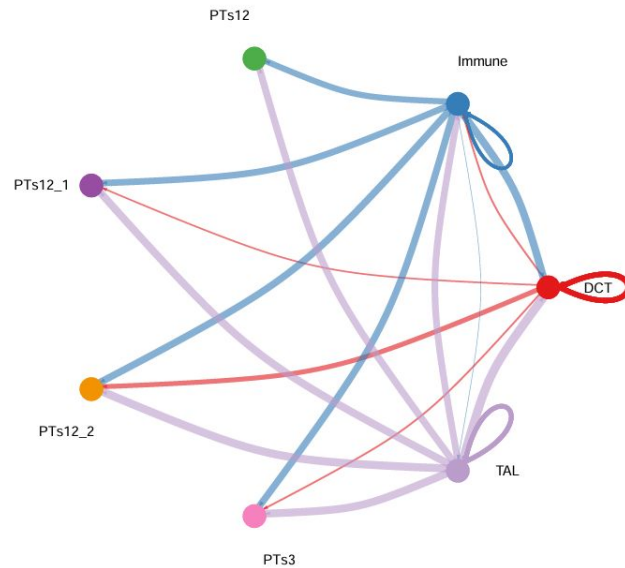


TGFb Signaling Pathway Network (Repair)

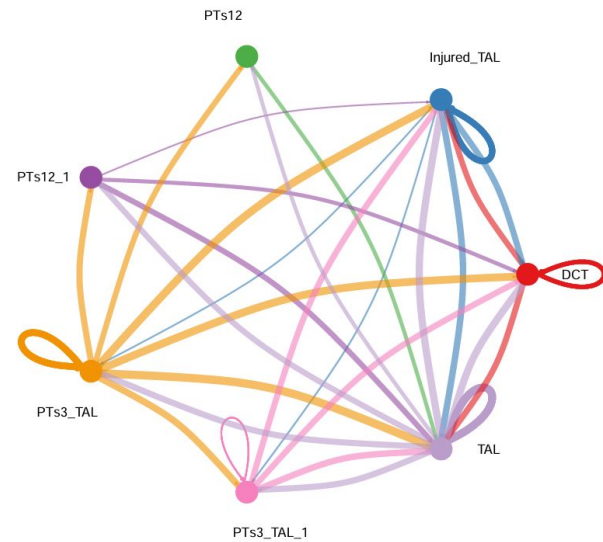
Sham



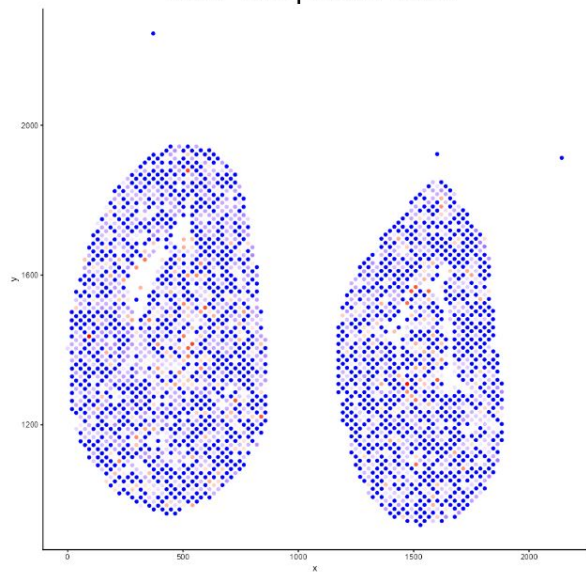
4 Hours



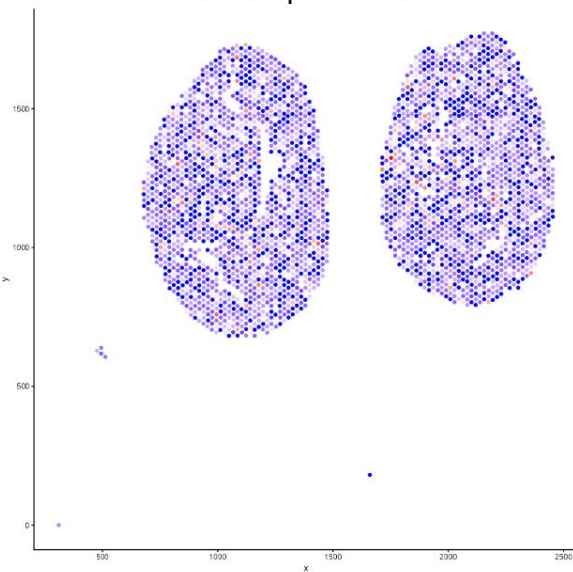
12 Hours



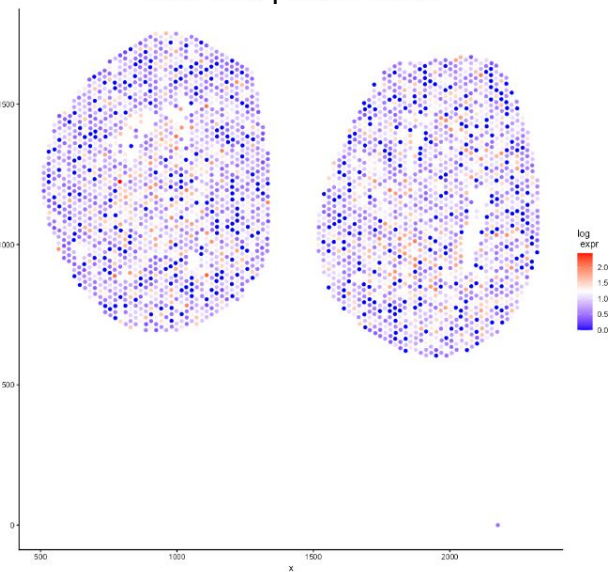
Tgfb1 - Sham
max=1.72 | mean=0.327



Tgfb1 - 4h
max=2.32 | mean=0.508

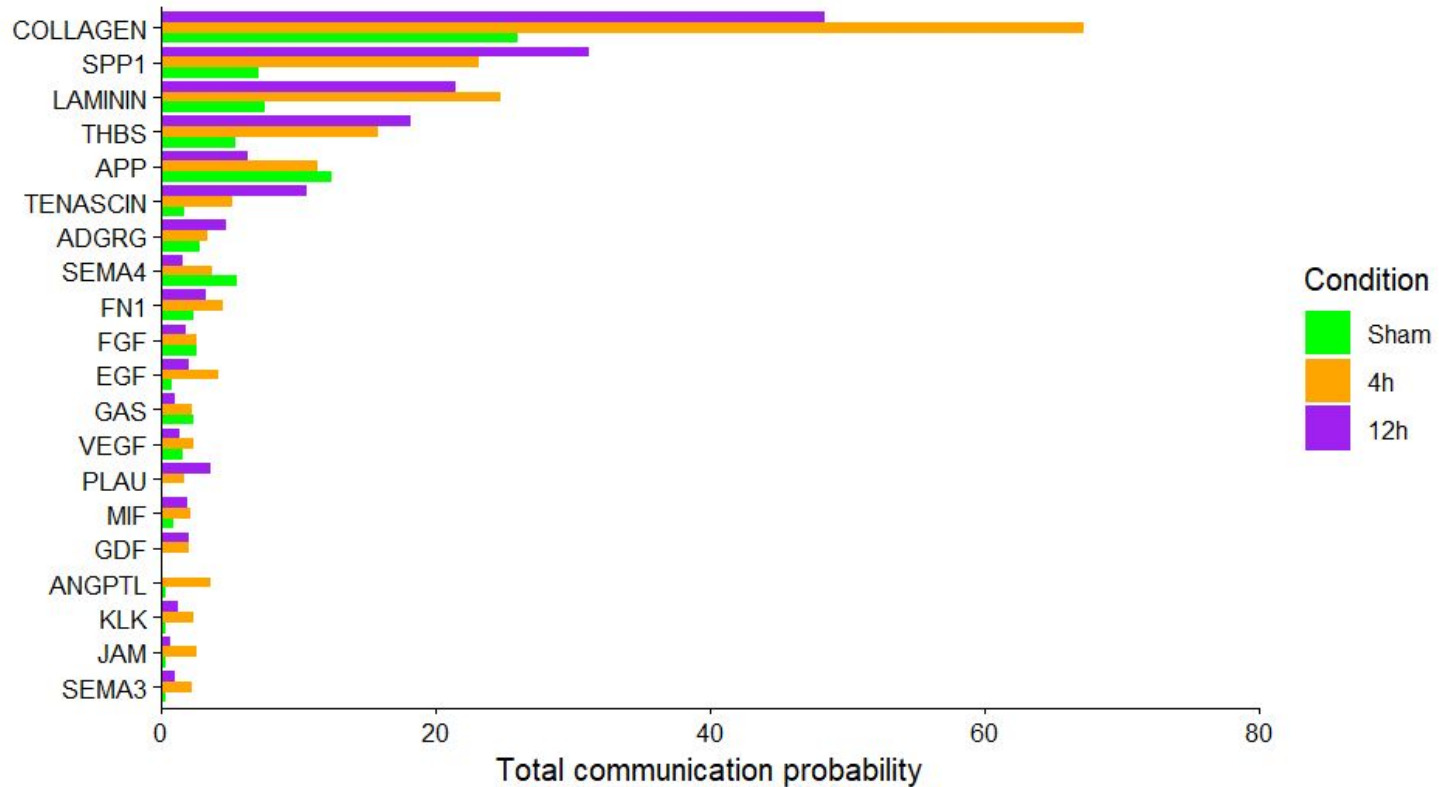


Tgfb1 - 12h
max=2.45 | mean=0.834



Pathway information flow across IRI timepoints

ECM repair and Immune Cell Recruitment Over Time

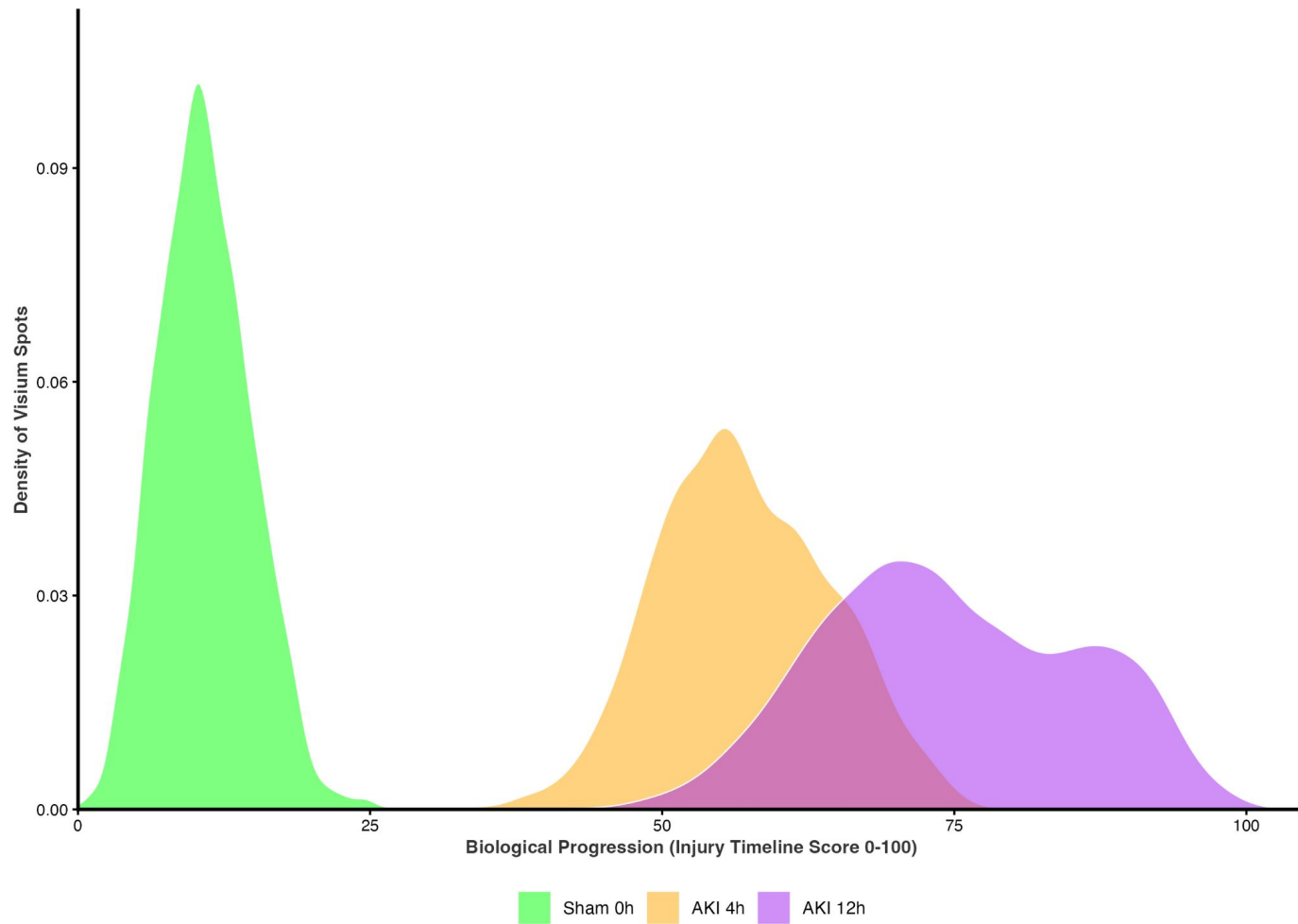


Pseudotime Analysis

A form of data analysis in which certain markers are identified as a means to label certain cell populations as being on a certain timeline relative to the rest of the population.

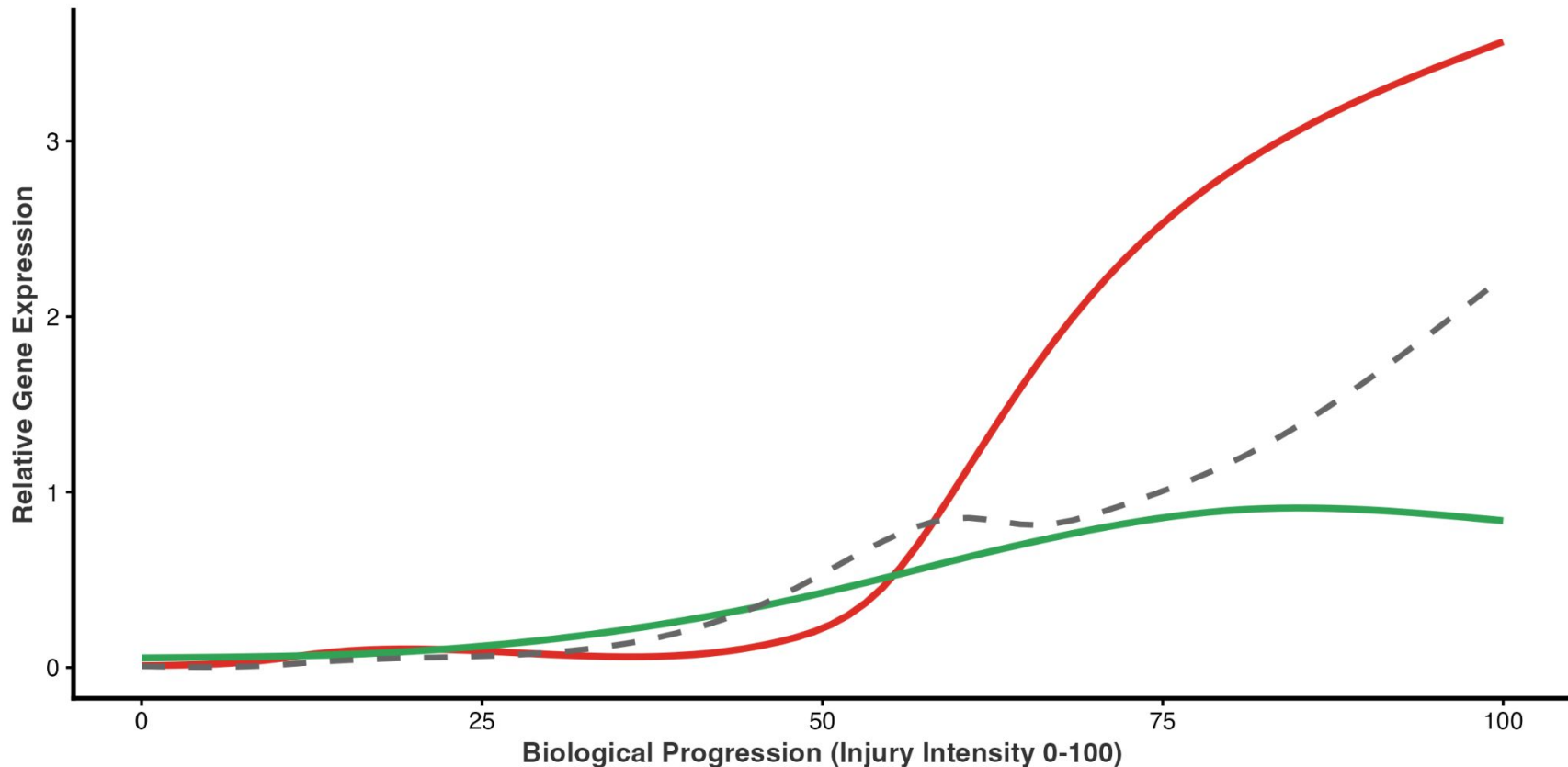
Alignment of Biological and Experimental Timelines

The sequential shift of density peaks proves our Injury Timeline accurately captures the experiment's progression.



Biological Progression of the Injury Signature

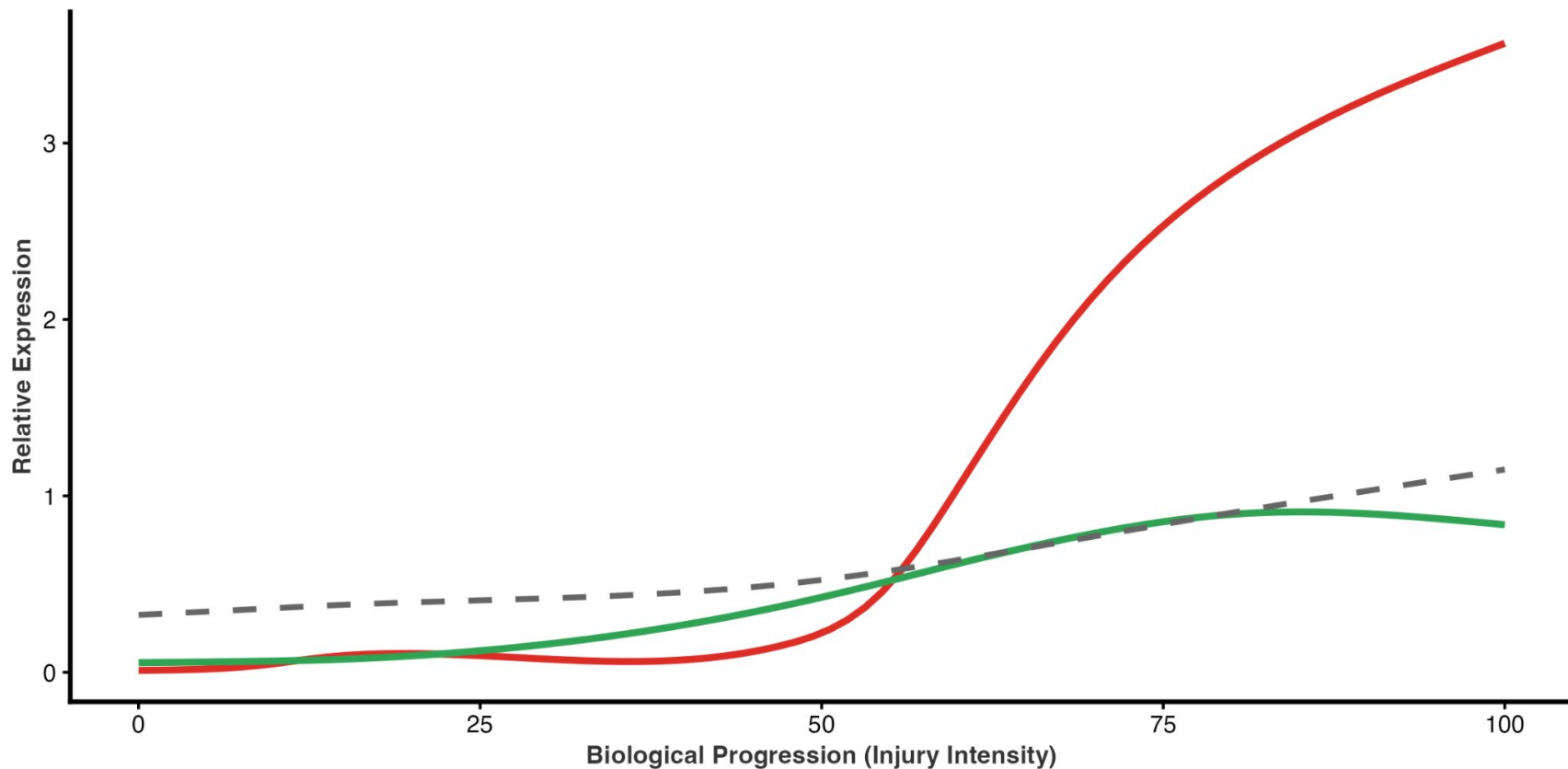
Timeline anchored to the discovery-driven injury signature reveals the competition between cell fate and scarring.



Biological Path — Acute Injury (Havcr1) — Cell Fate (Sox9) - - - Fibrotic Signaling (Serpine1)

Competitive Signaling of Regenerative and Fibrotic Programs

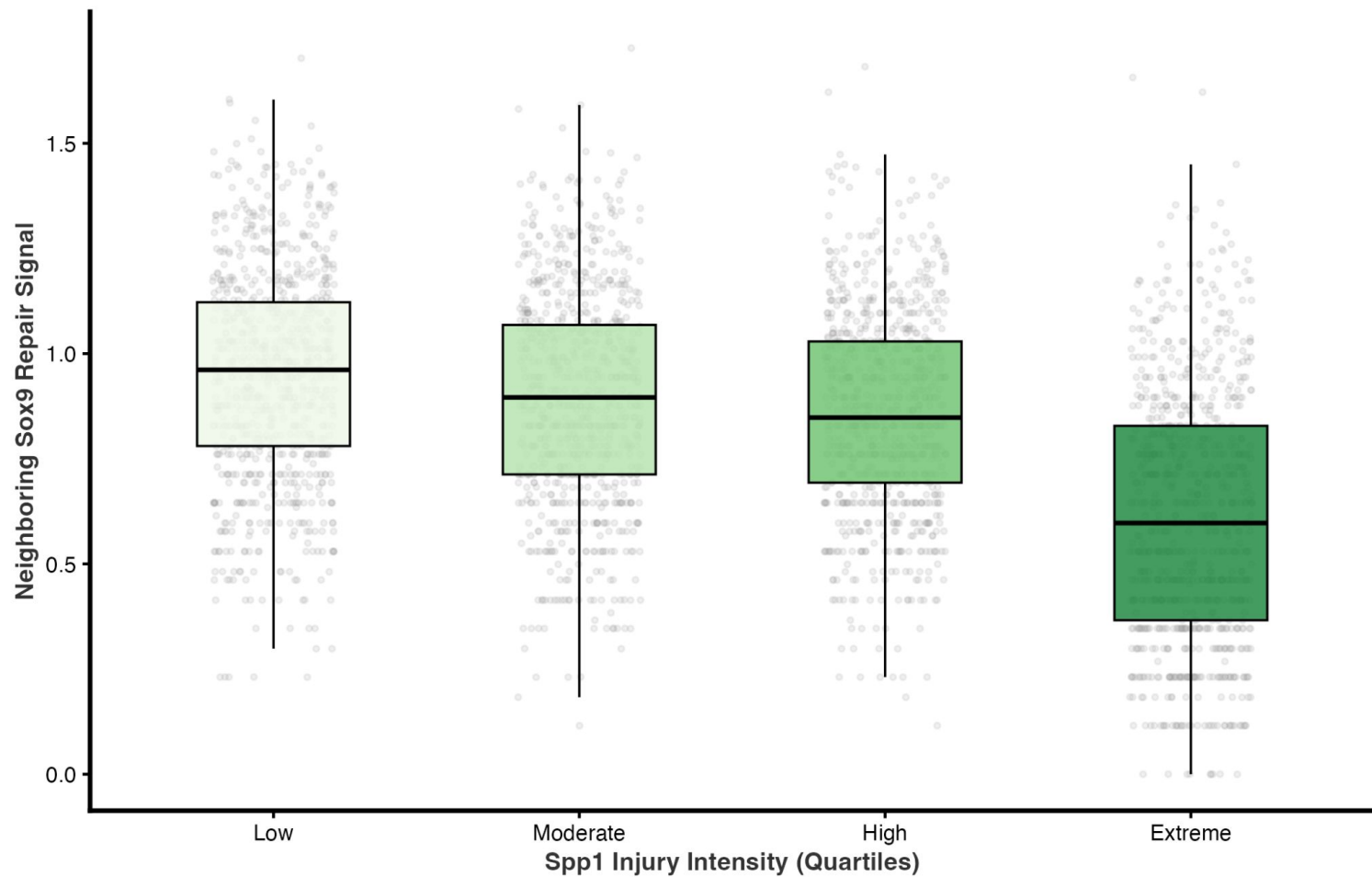
At peak injury, the regenerative cell fate program is caught in a conflict with rising remodeling signals.



Biological Path — Acute Injury (Havcr1) — Cell Fate (Sox9) - - - Tissue Remodeling / Repair (TGF-Beta)

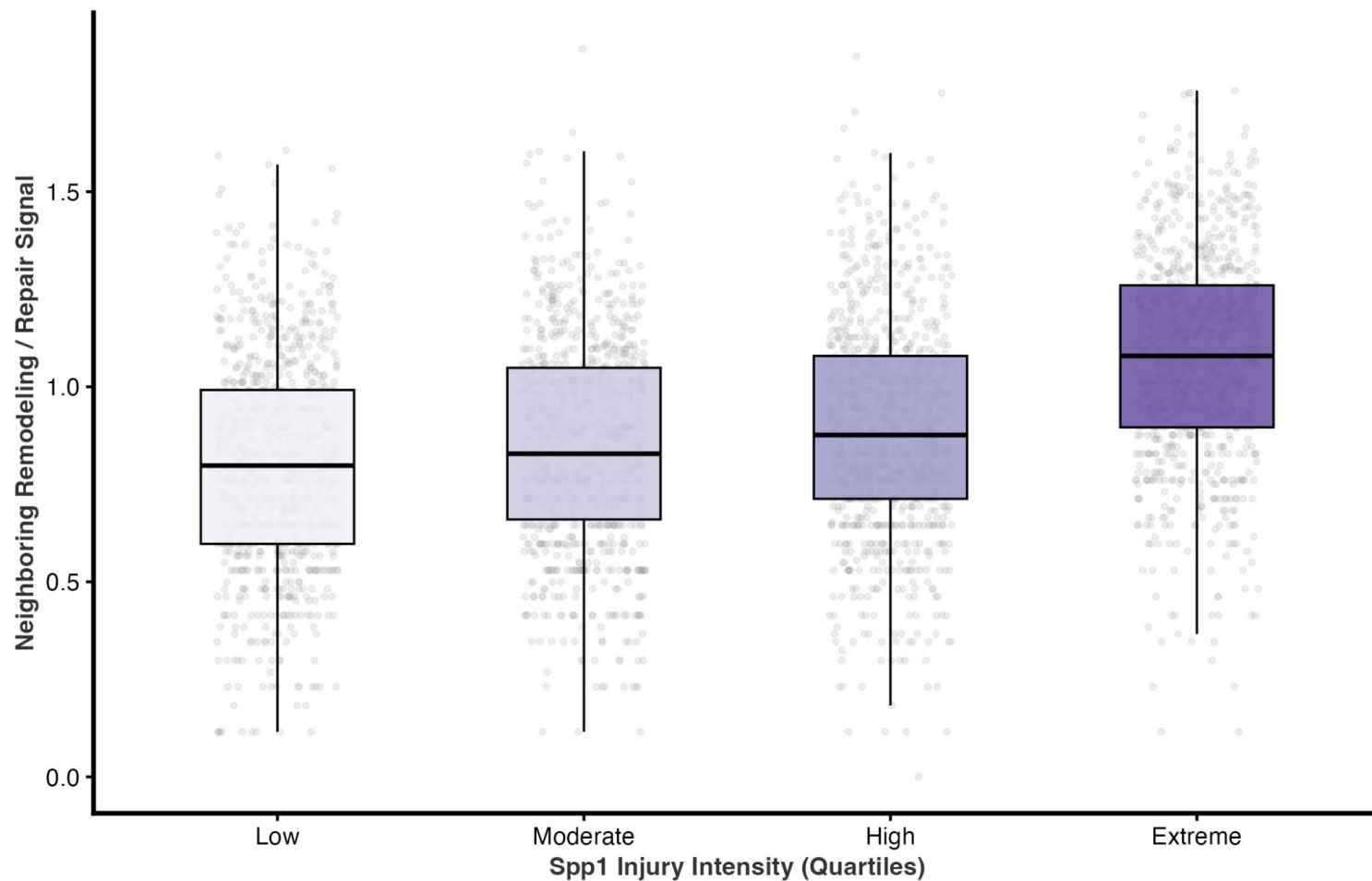
Spatial Association: Damage vs. Repair Signaling

Neighboring Sox9 repair signals seem to decrease as localized Spp1 injury intensity rises.



Spatial Association: Damage vs. Tissue Remodeling

Neighboring remodeling signals (TGF-Beta) increase proportionally as localized Spp1 injury rises.

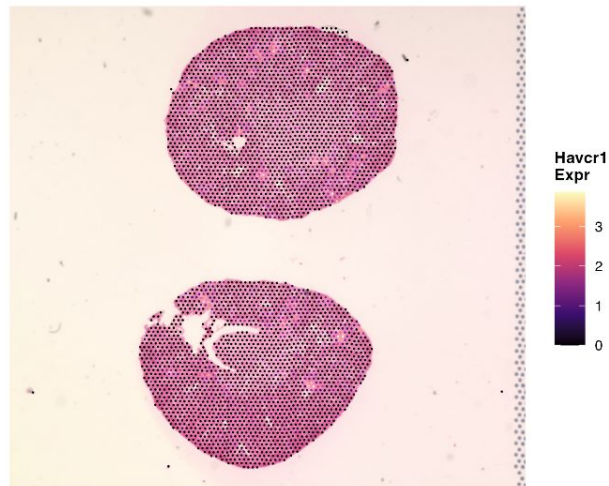


Validating the Trajectory: The AKI-to-CKD Transition at 6 Weeks

Our 12h pseudotime model successfully predicted the failure of regeneration and the establishment of chronic fibrosis.

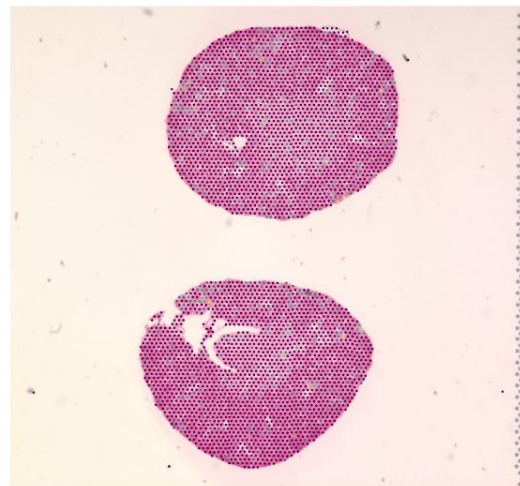
1. The Ghost of Injury

Acute damage markers (Havcr1) have resolved at 6 weeks.



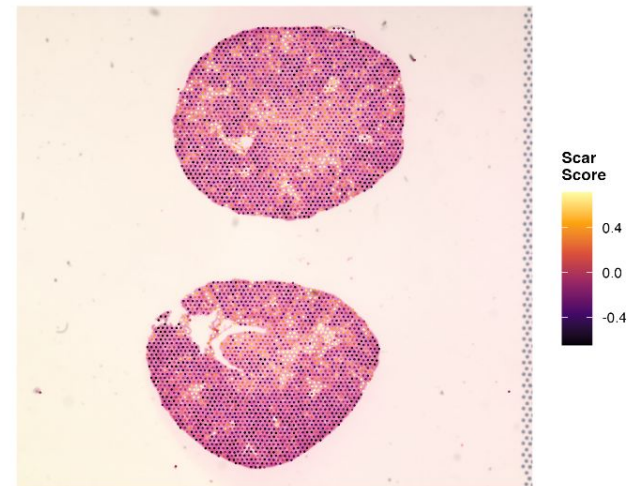
2. The Abandoned Fate

The S3 regenerative program (Sox9) failed to establish.



3. The Permanent Scar

Fibrotic signaling permanently occupies the predicted cortical niches.



Biological Roadmap

At 4 hours after IRI, the cortical proximal tubule cells start to express injury markers.

By 12 hours, the thick ascending limb cells in the cortex-medulla boundary joins the injured population and begin send a ton of recruitment signals [SPP1 (osteopontin), CCL6, CXCL12, IL34], and macrophages start accumulating in the CMJ

Injured tissue is simultaneously initiating repair (Sox9, Tgfb), but the fibrotic program is escalating faster.

Limitations

- Visium resolution (55 μ m spots) means each spot contains many cells. This creates mixed signals where *macrophages* and *epithelial* cells are captured together.
- No pure macrophage topic emerged in topic 12 because macrophages are always intimately associated with injured tubules rather than forming discrete clusters. This is biologically interesting but technically limiting.
- Dominant-label assignment for CellChat (which.max on theta) discards the continuous mixture information that LDA preserved. A spot gets collapsed to a single label, potentially losing real co-localization signal.
 - Miss signaling happening between co-localized cell types within the same spot
 - Misattribute signals to the dominant cell type when a minority population is actually the sender/receiver



References

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Acknowledgment

All Dedenne Images were created with Gemini Nano Banana

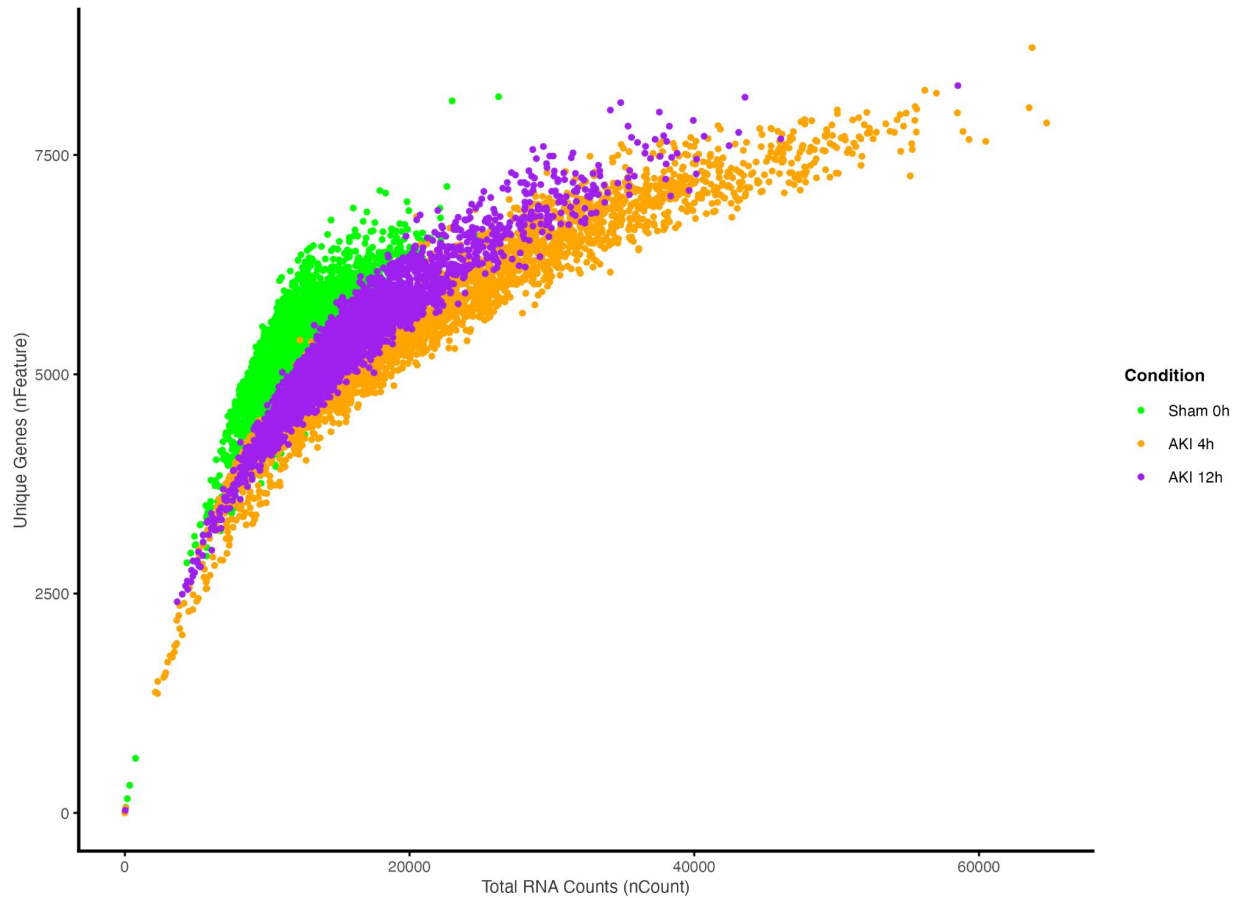
All Lillian code is on:

https://github.com/Lillian-Lam/Omic-Projects/tree/main/Genomic_Data_Visualization_EN.580.428

Supplementary

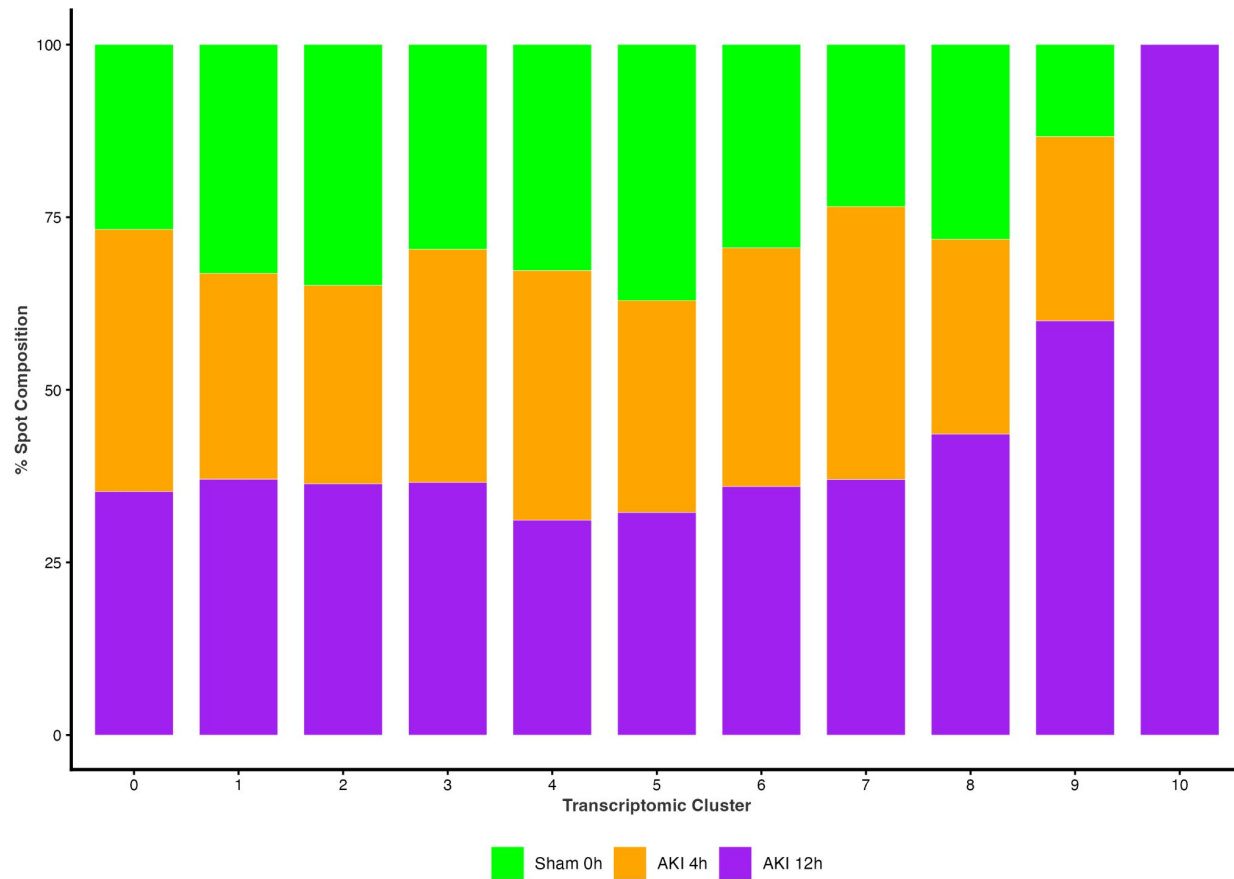
Quality Control: Transcriptional Complexity

Consistent linearity between RNA counts and unique genes confirms technical integrity across the 12-hour injury timeline.



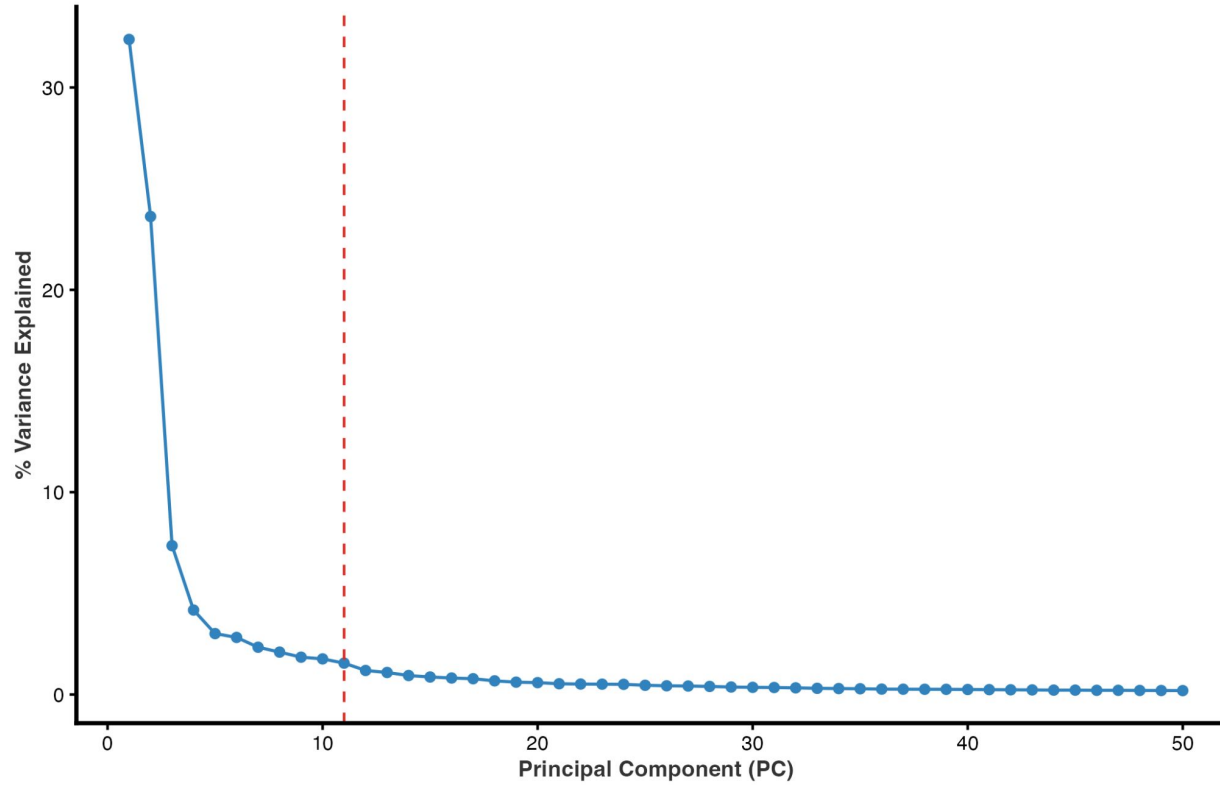
Harmony Integration: Cluster Mixing Analysis

Cluster-wise composition confirms successful technical alignment, isolating Cluster 10 as a unique 12h injury state.



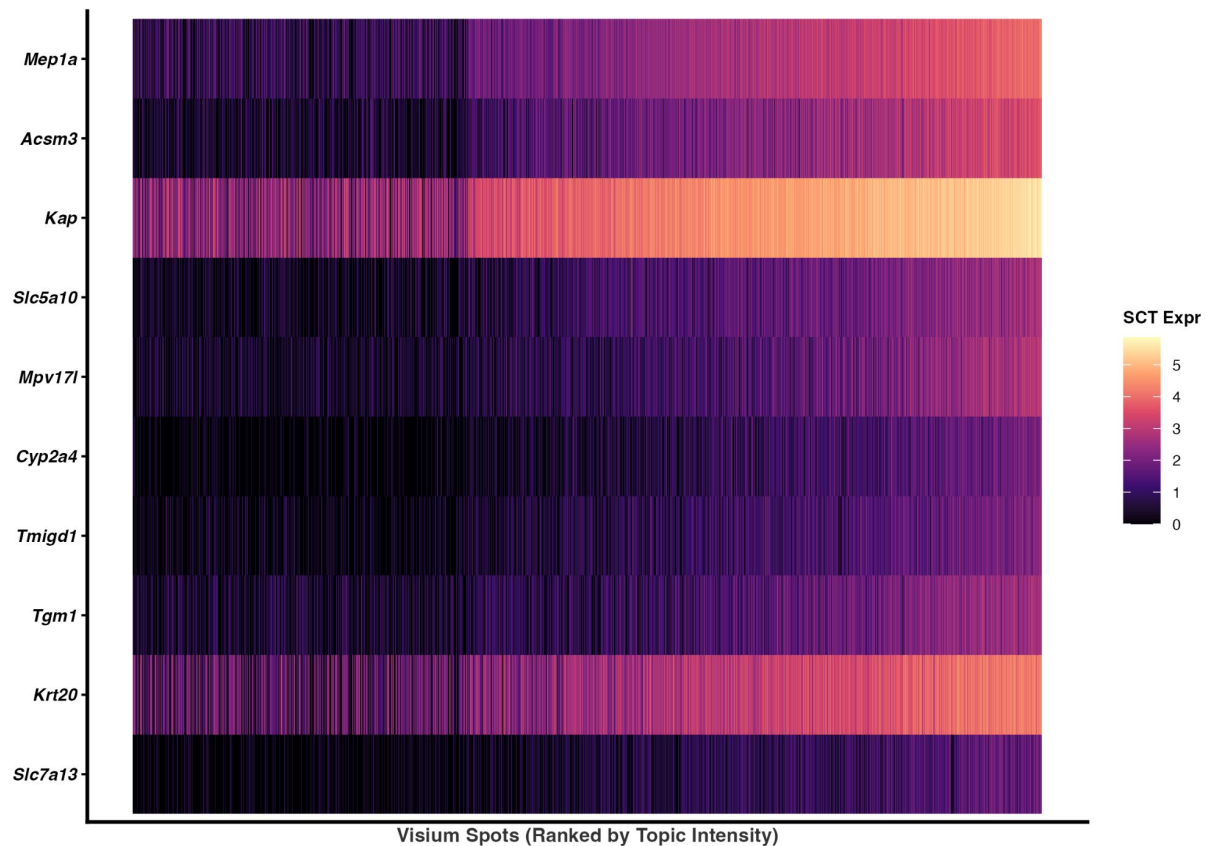
PCA Elbow Plot: Dimensionality Selection

The inflection point at PC 11 confirms the capture of core biological variance while excluding technical noise.

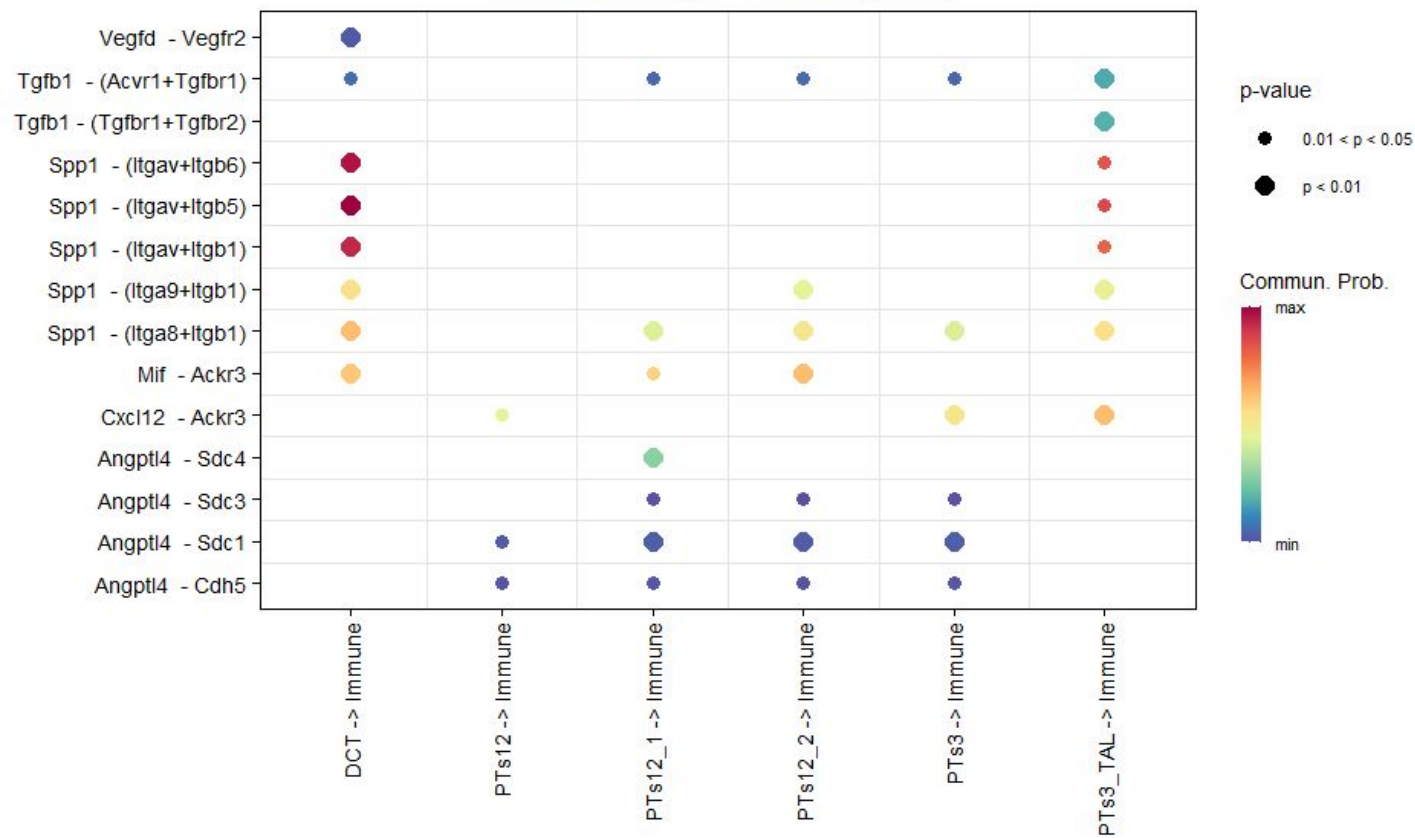


Cellular Identity: Topic 5 Markers

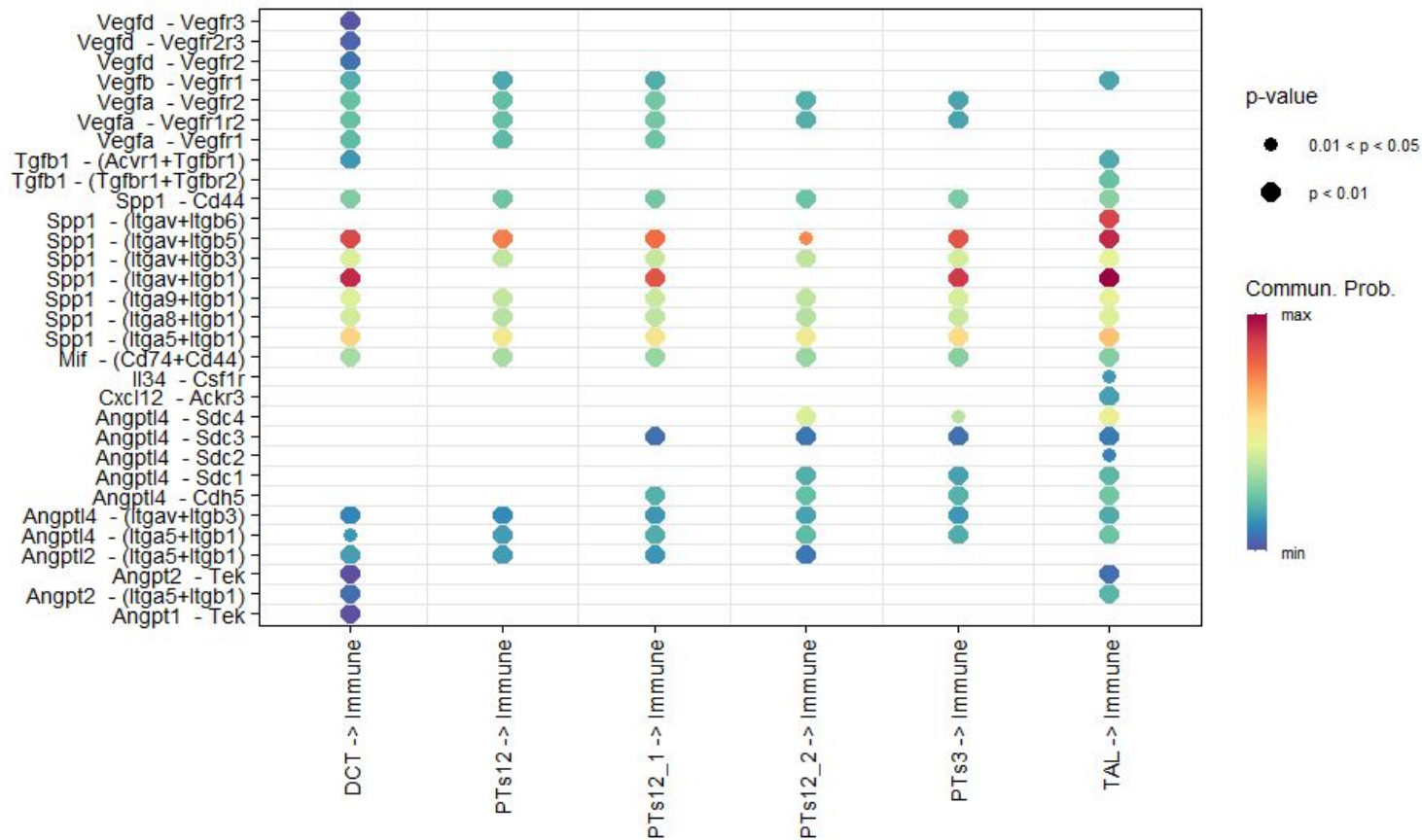
S3 segment marker expression increases linearly with Topic 5 proportion, confirming its identity.



Recruitment signals Immune (Sham)

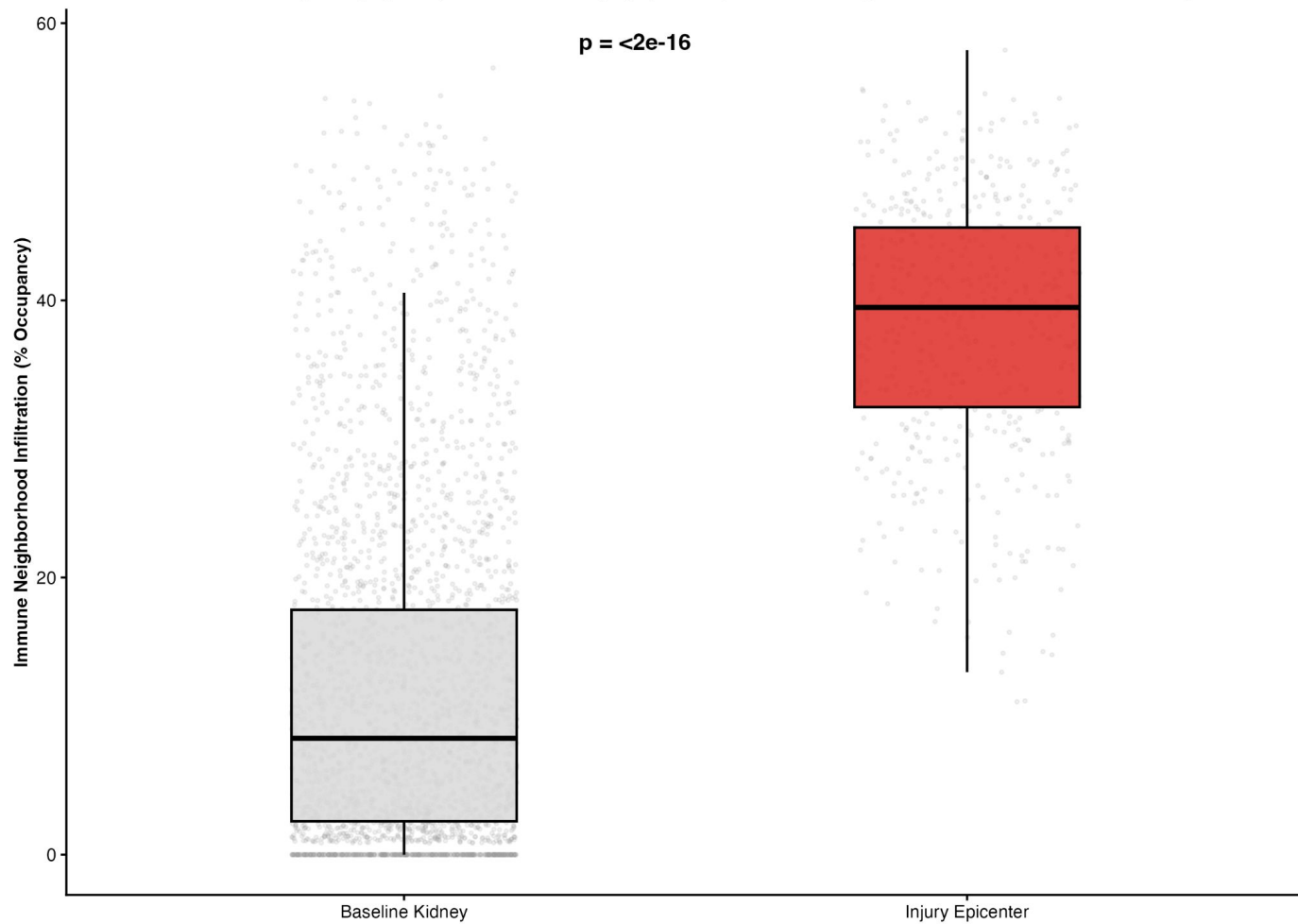


Recruitment signals Immune (4h)

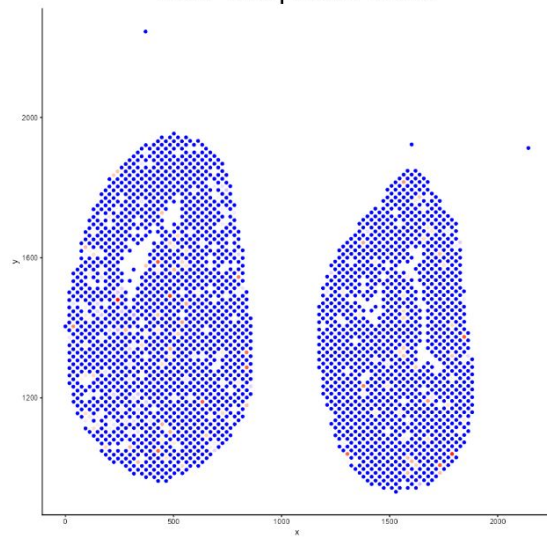


The Threshold of Invasion: Biological Trigger Point

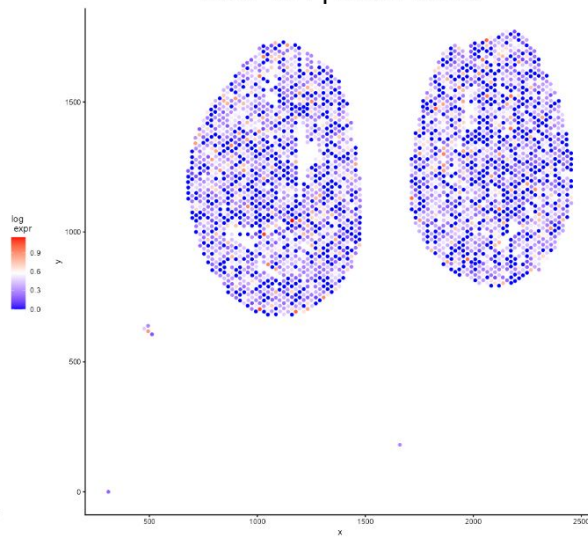
The immune system physically 'colonizes' the injury epicenter, while remaining at baseline in the rest of the kidney.



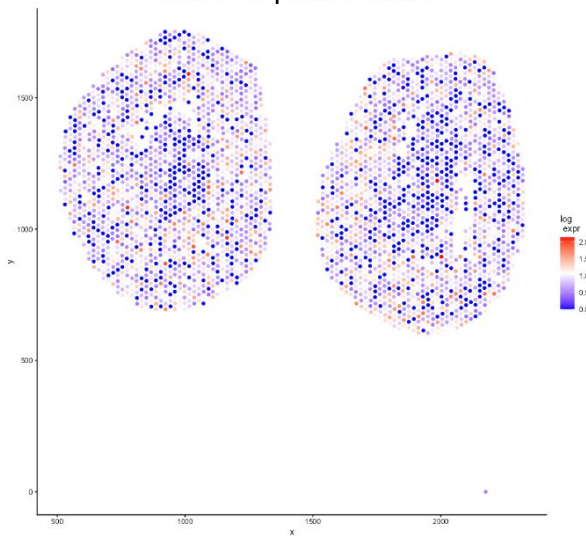
Sox9 - Sham
max=1.15 | mean=0.058



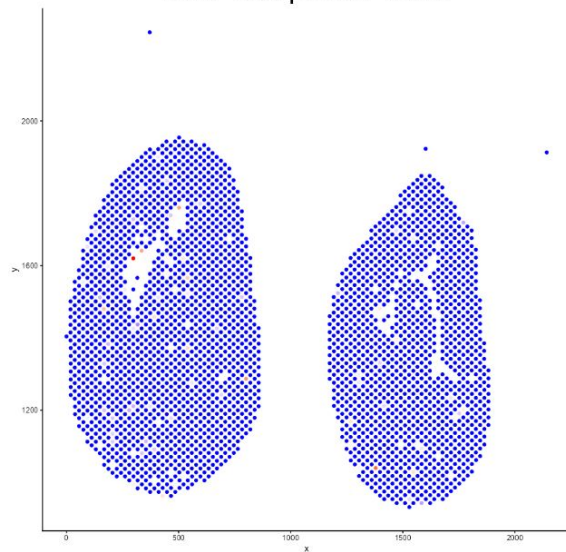
Sox9 - 4h
max=1.91 | mean=0.476



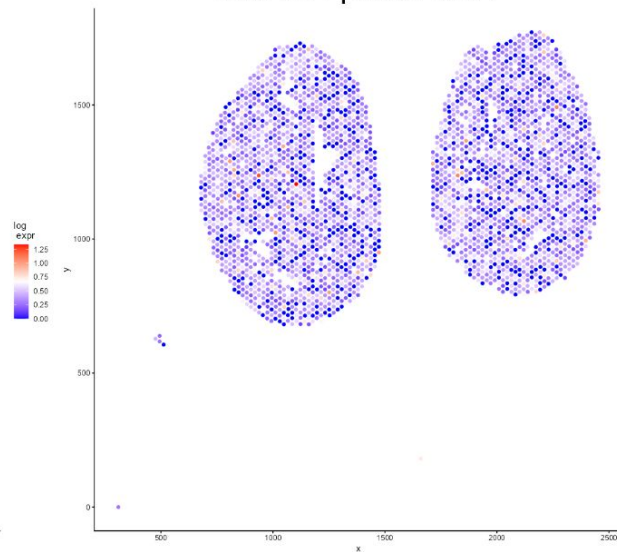
Sox9 - 12h
max=2.14 | mean=0.738



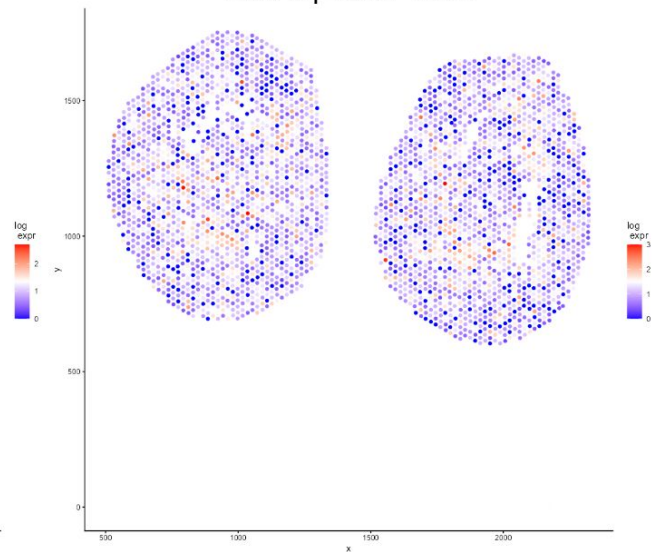
Serpine1 - Sham
max=1.34 | mean=0.019



Serpine1 - 4h
max=2.71 | mean=0.677

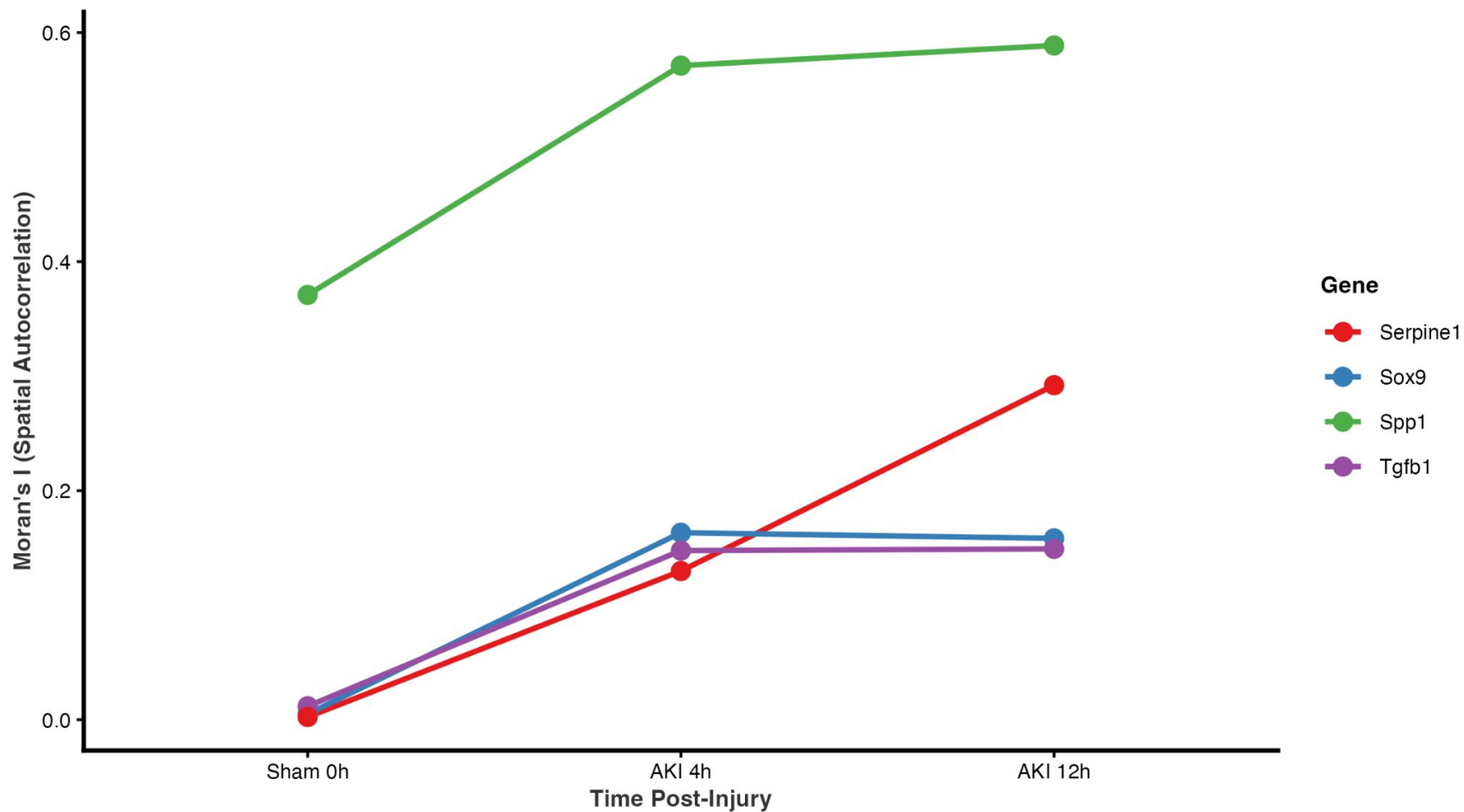


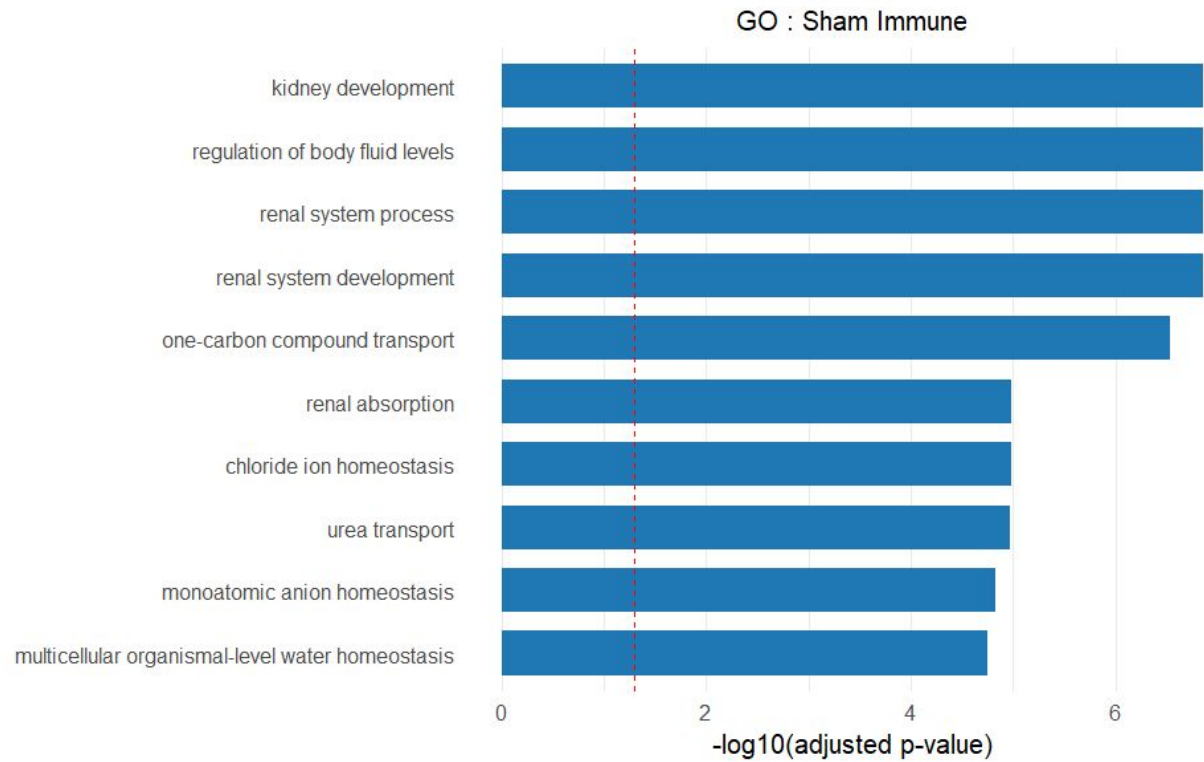
Serpine1 - 12h
max=3 | mean=1.003



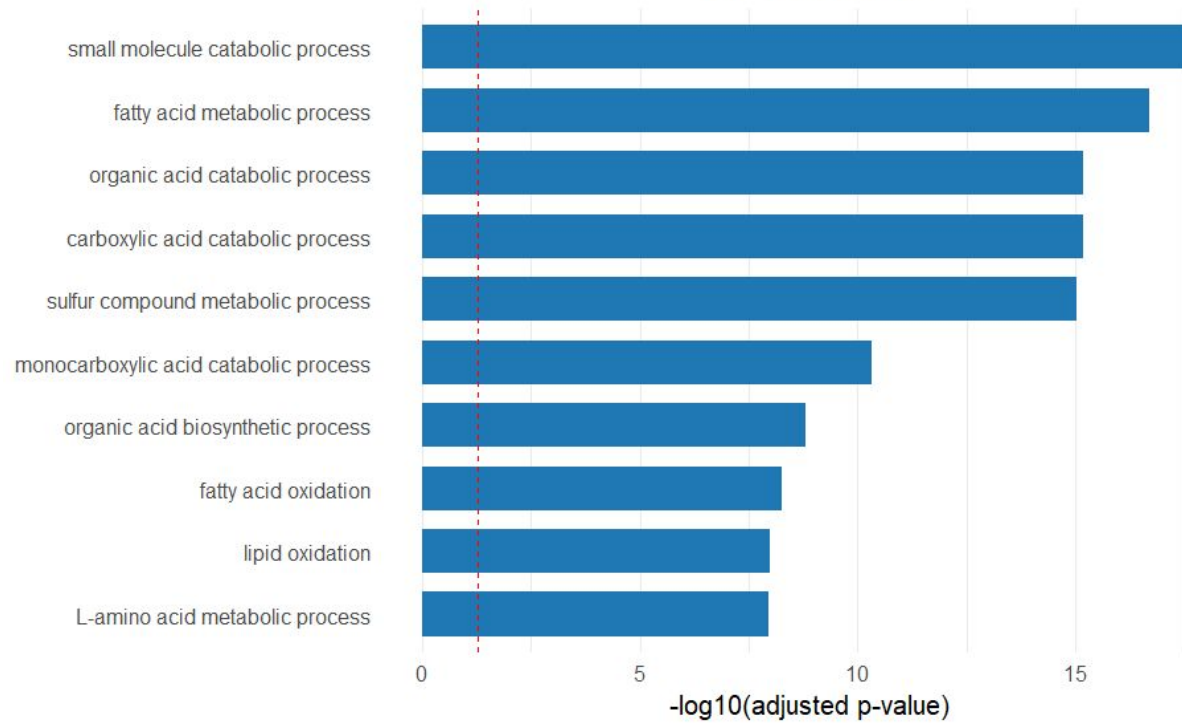
Spatial Organization Dynamics (Moran's I)

Rise in Moran's I values indicates the physical establishment of organized injury and repair niches.

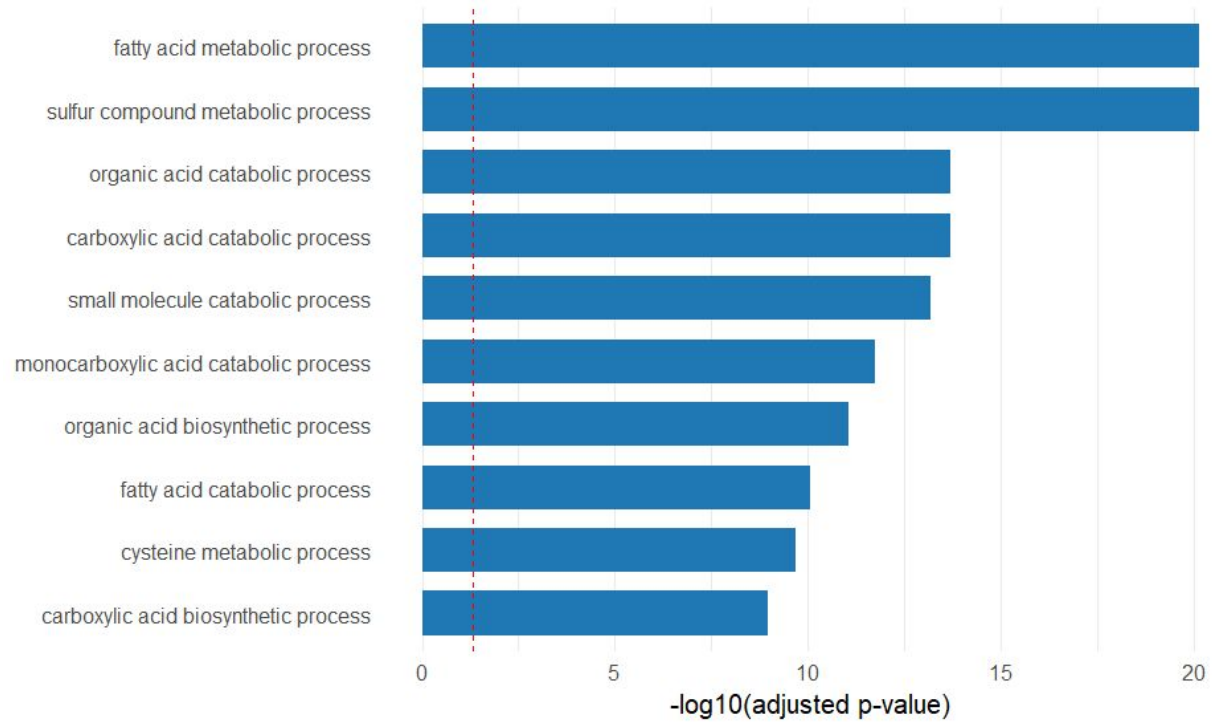


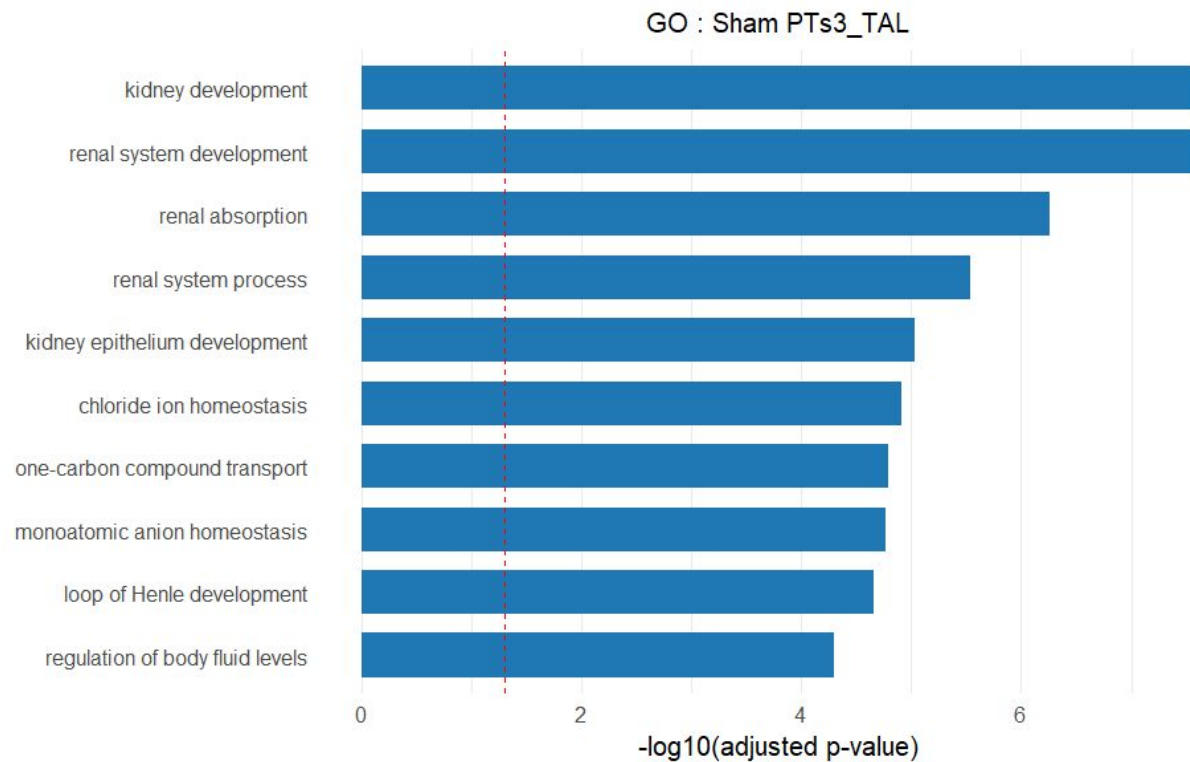


GO : Sham PTs12

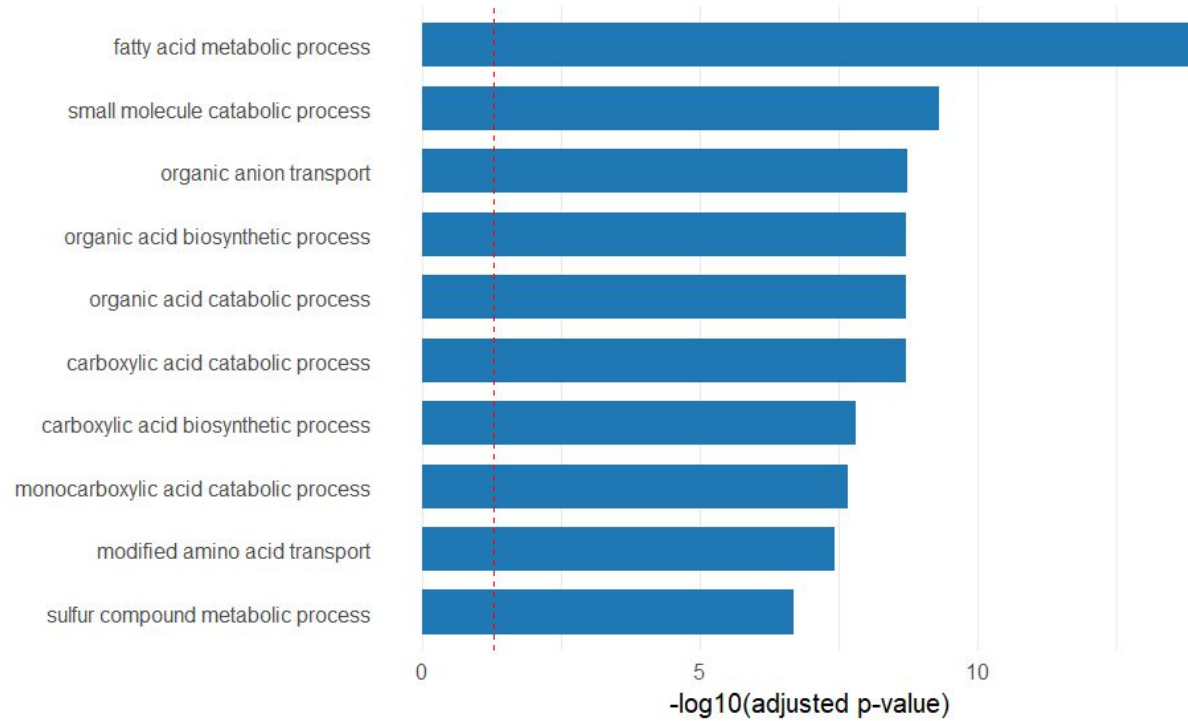


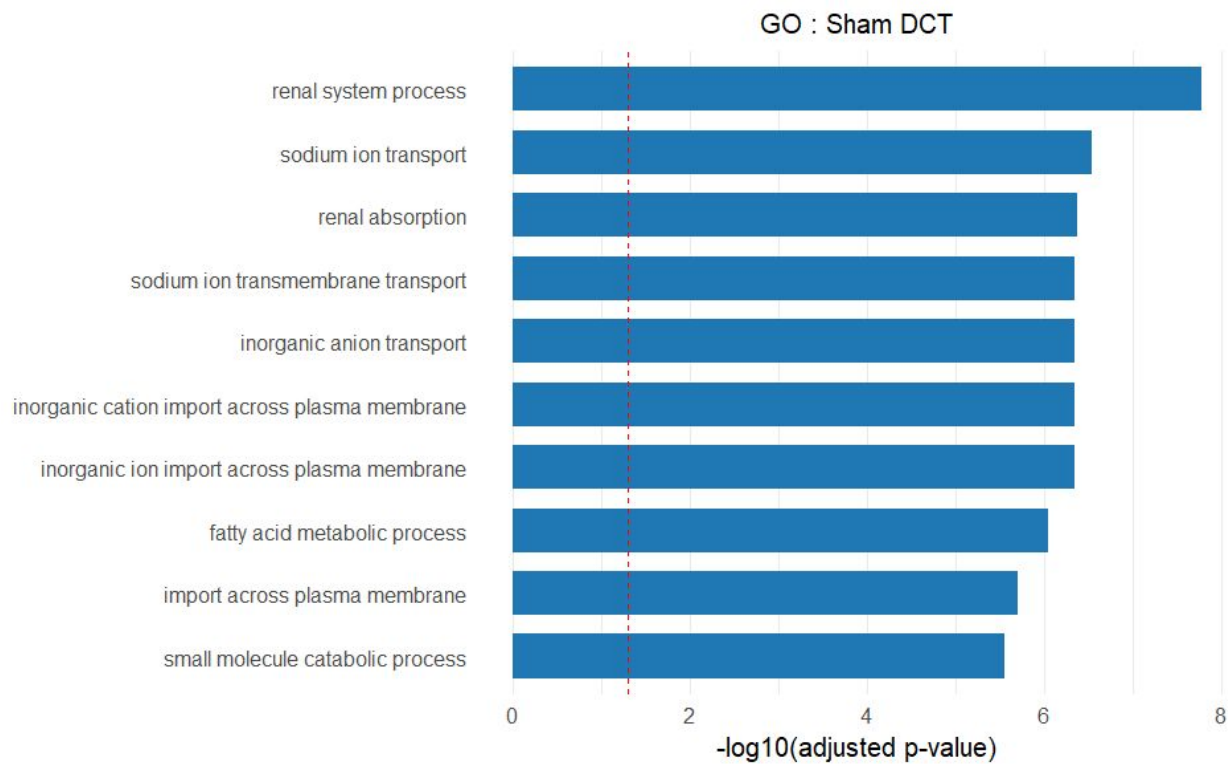
GO : Sham PTs3



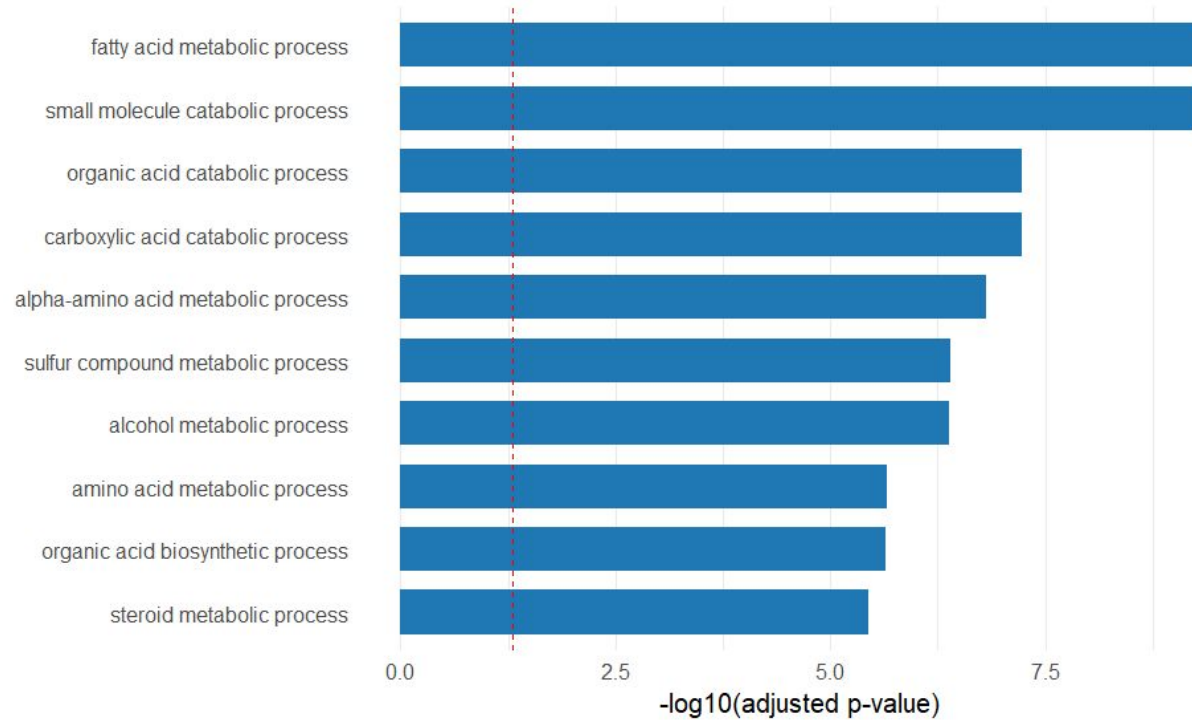


GO : Sham PTs12_1

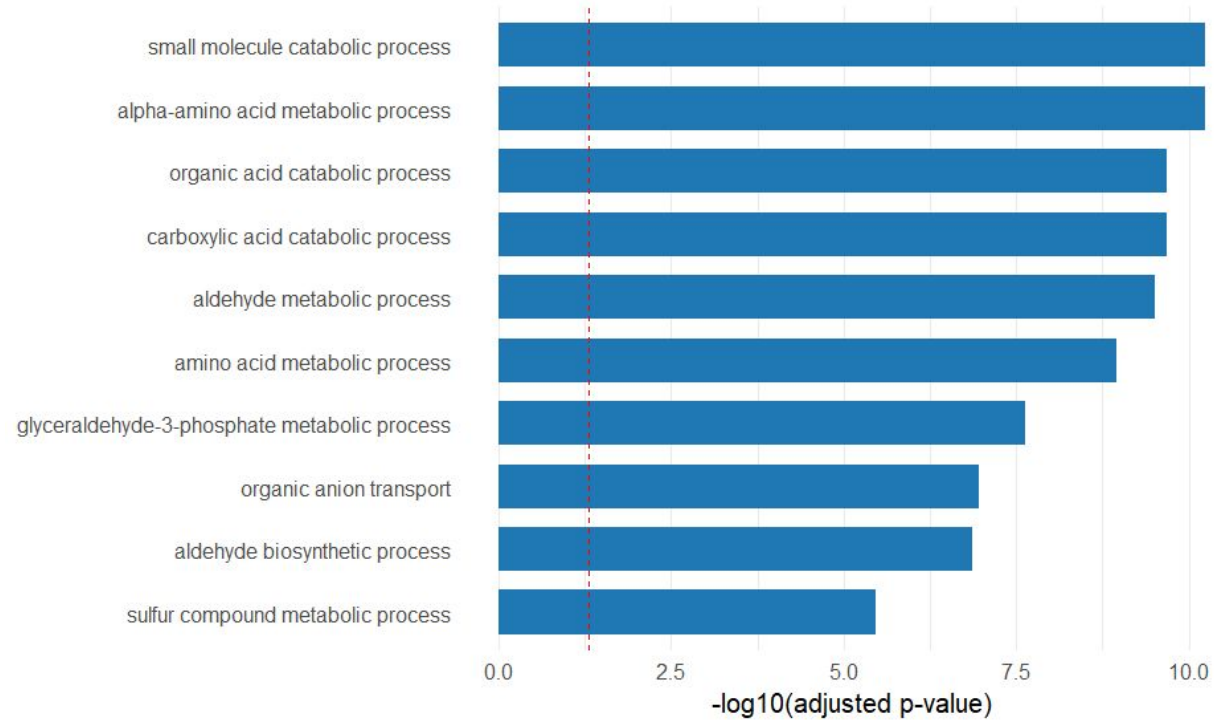


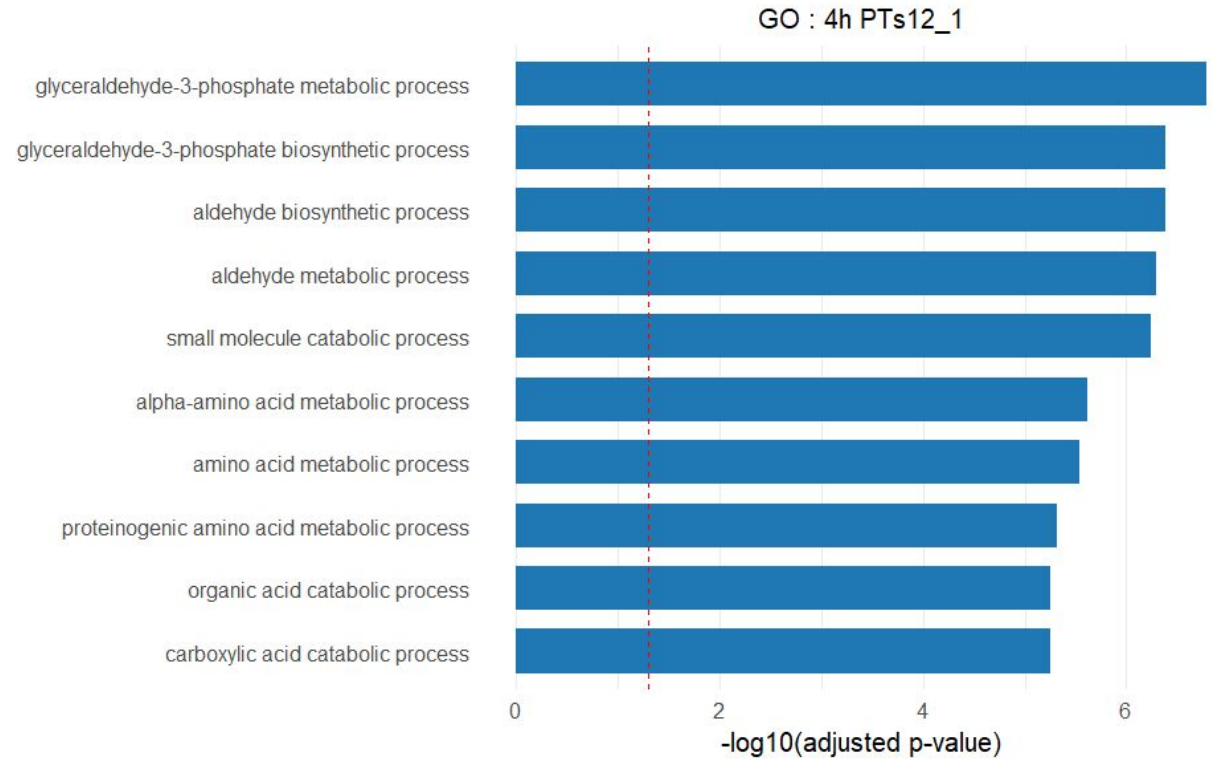


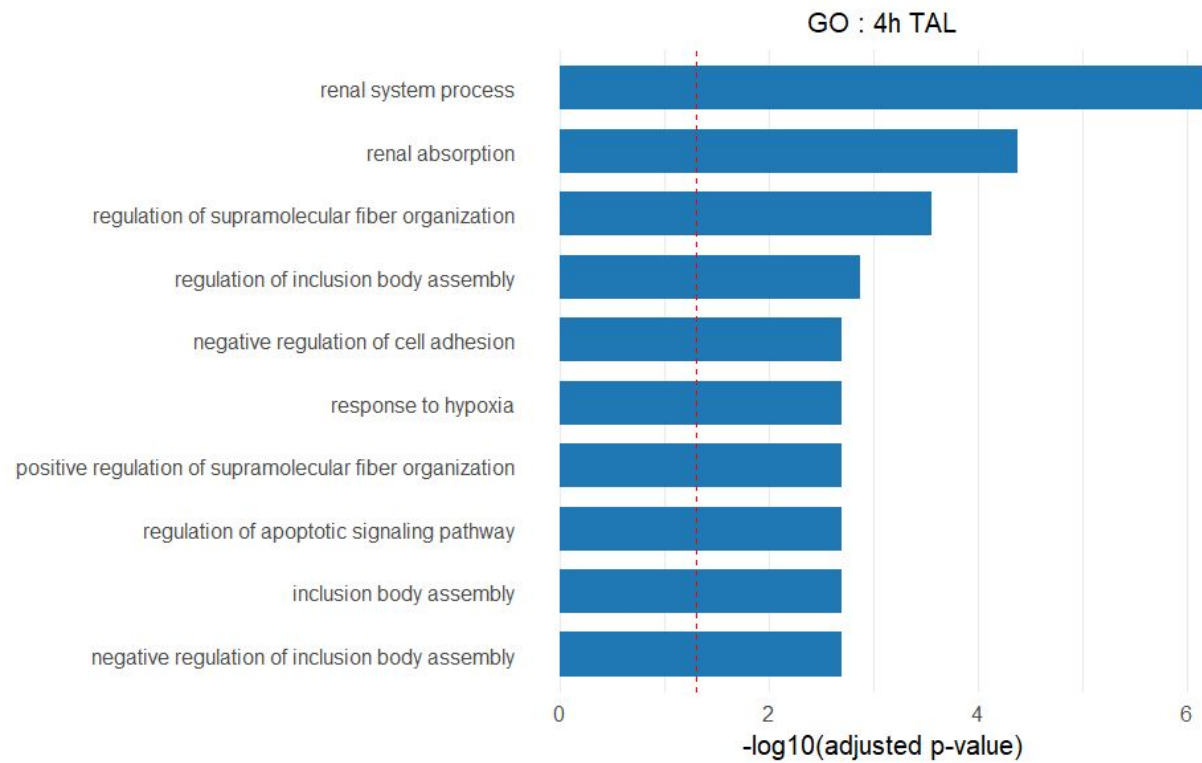
GO : Sham PTs12_2



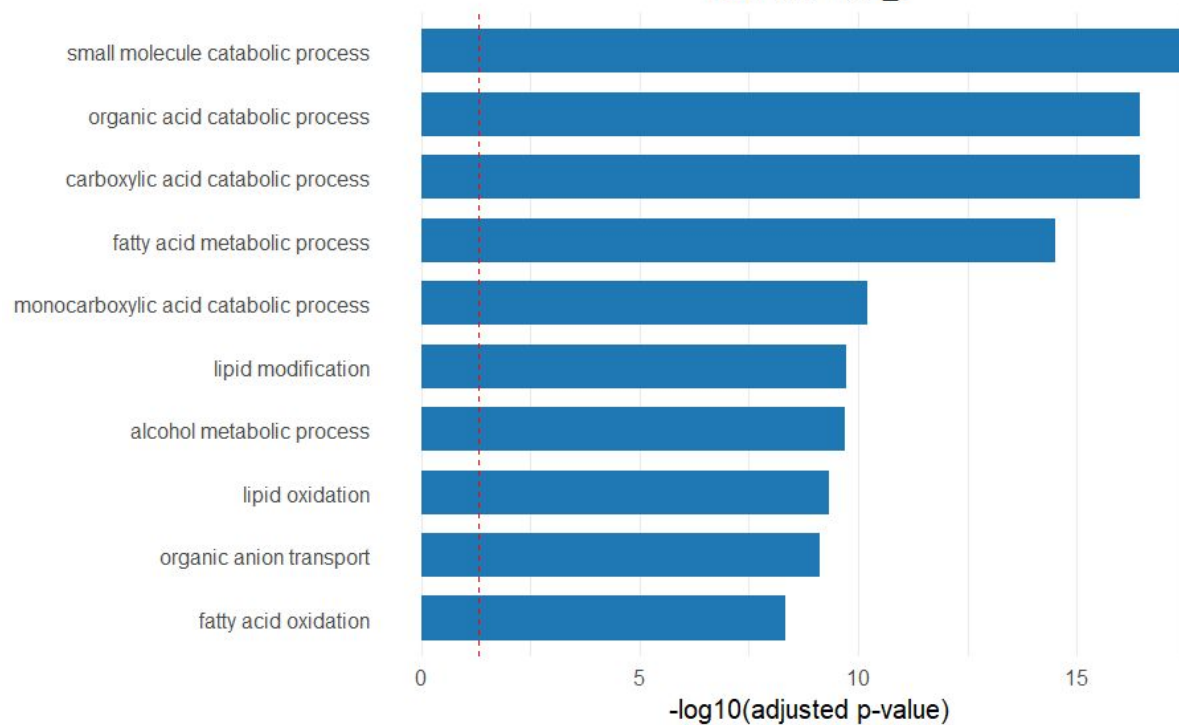
GO : 4h PTs12



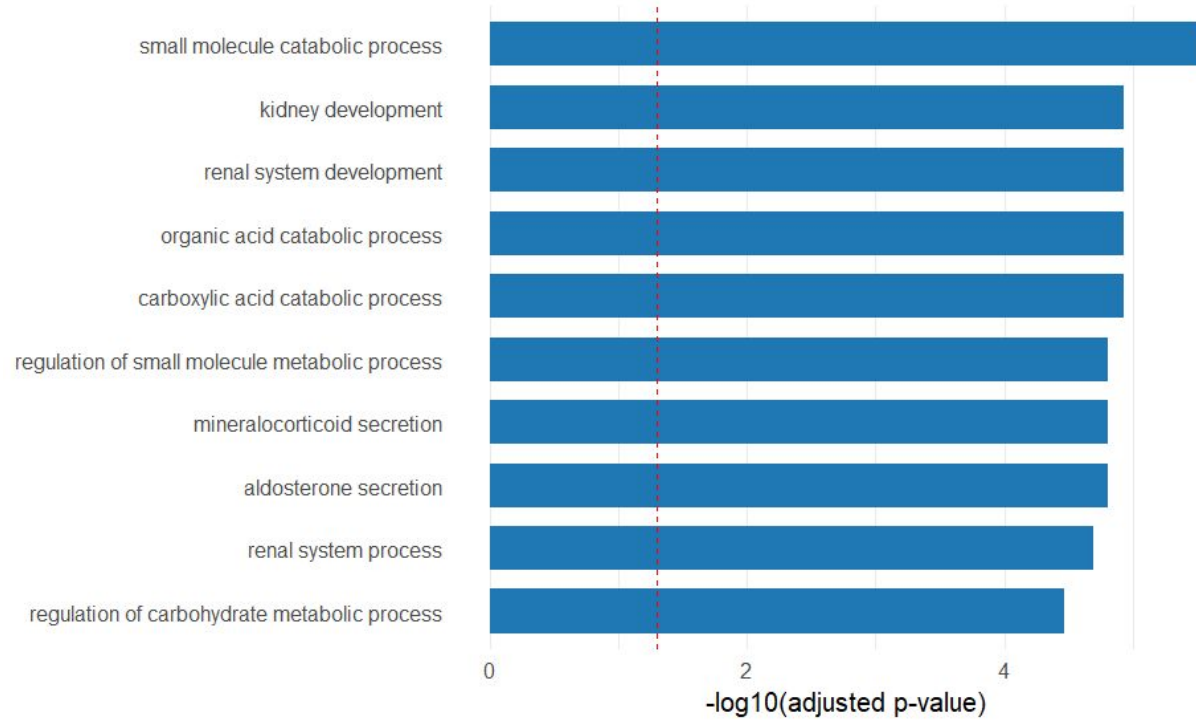




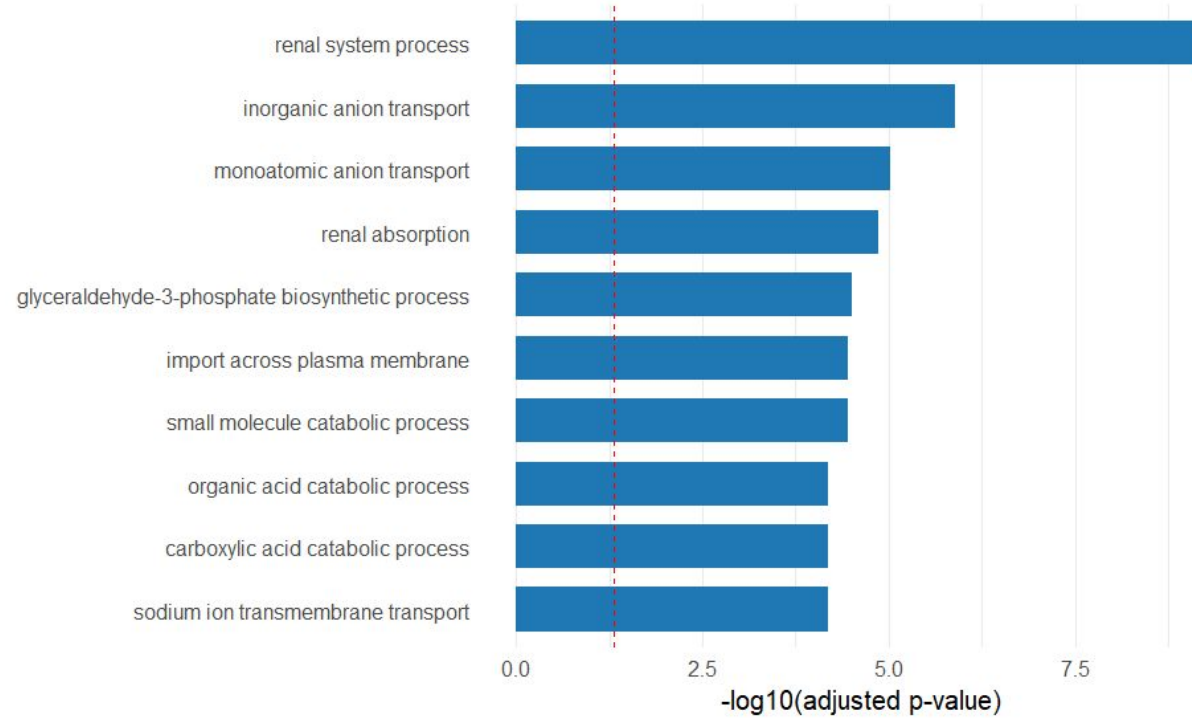
GO : 4h PTs12_2



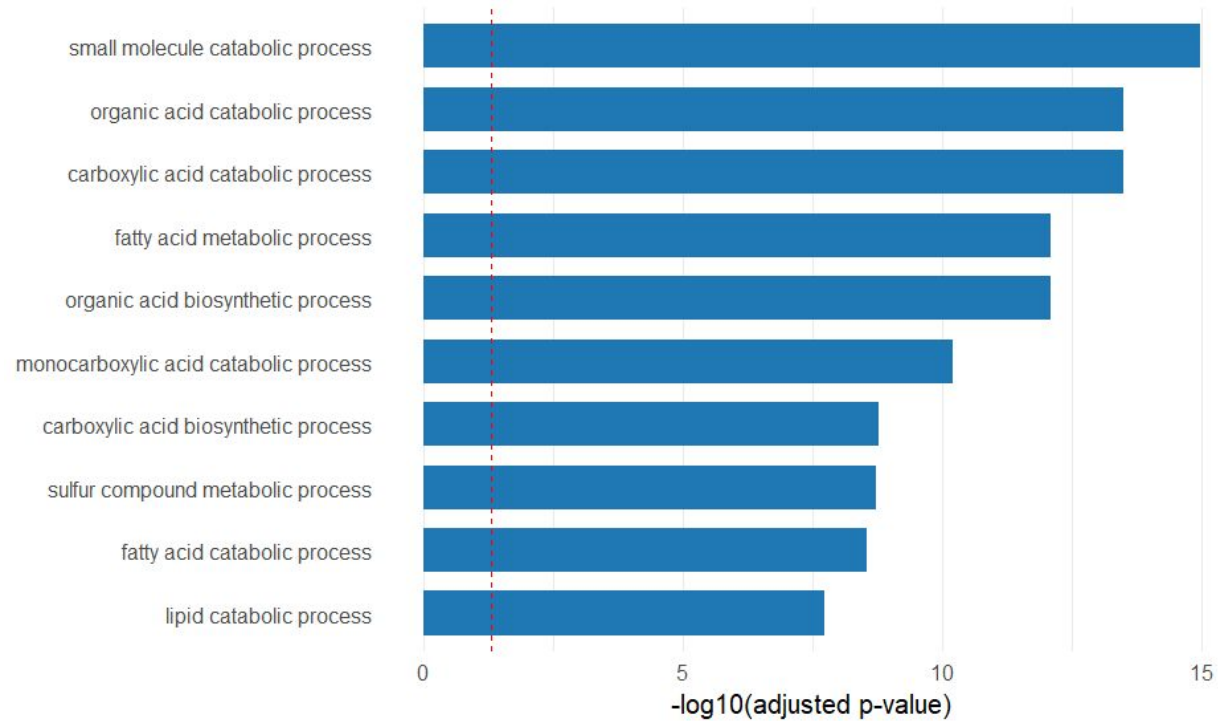
GO : 4h Immune

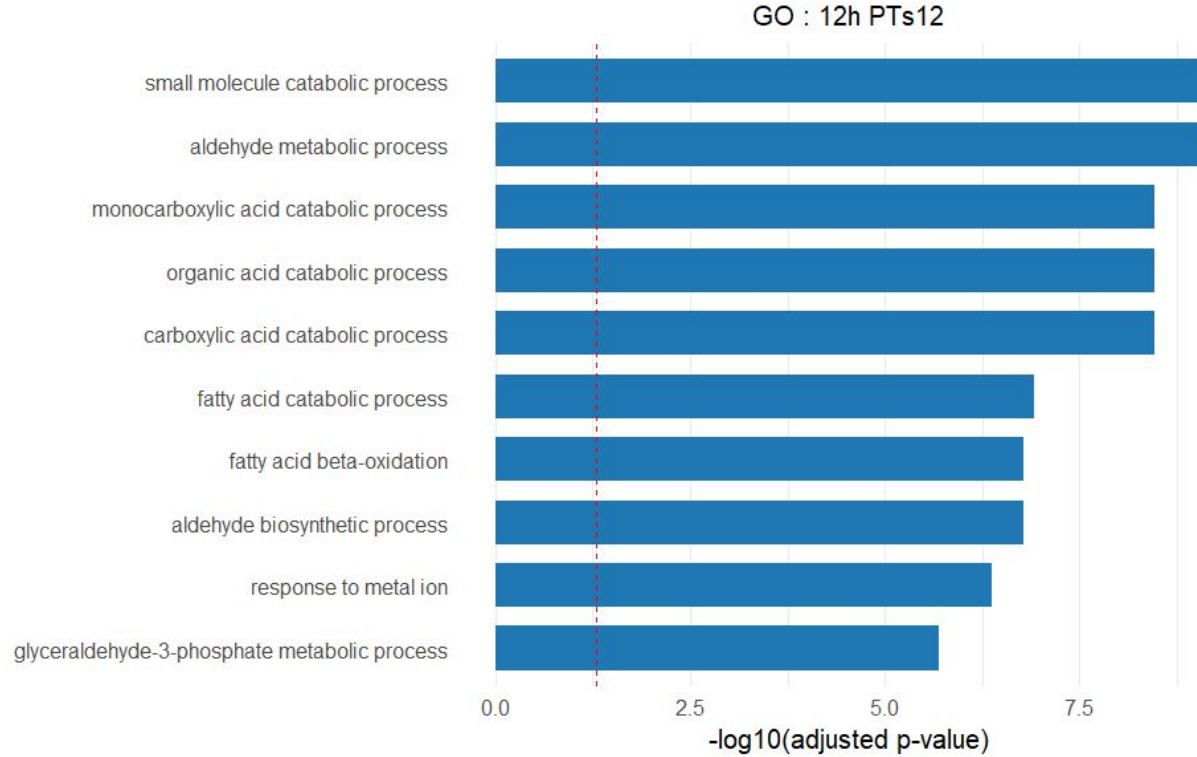


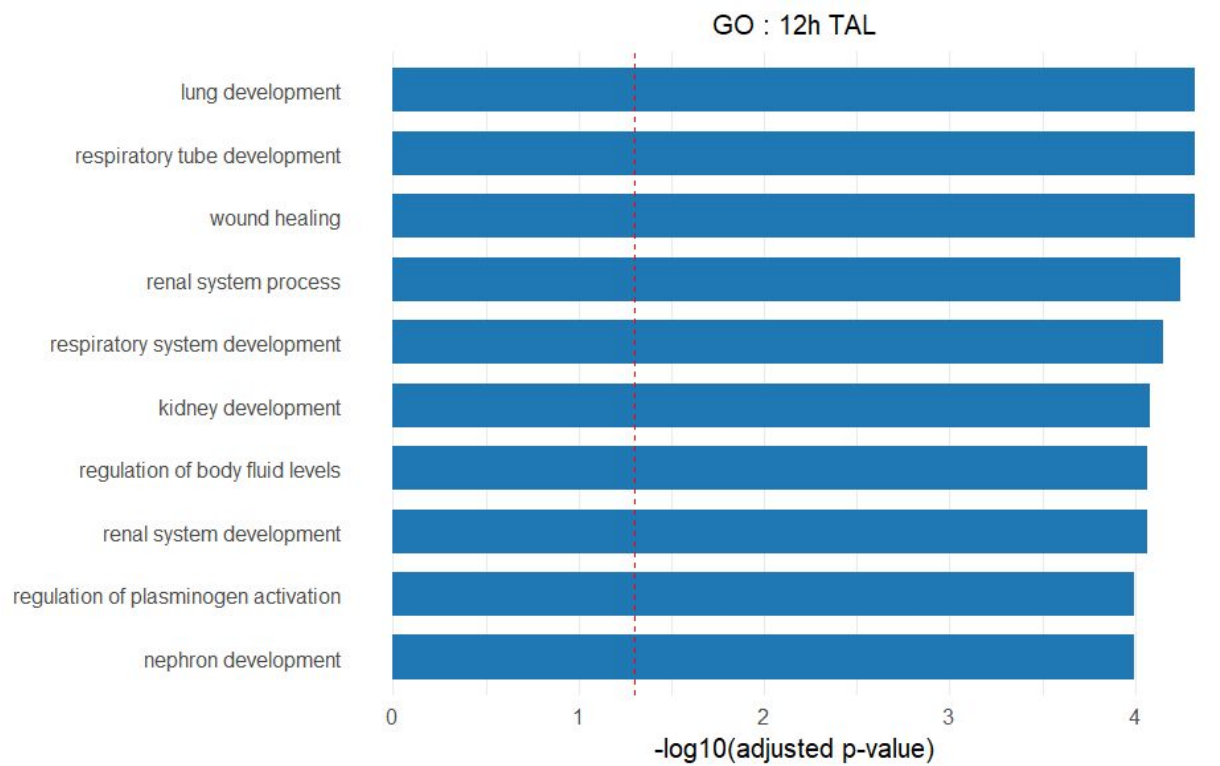
GO : 4h DCT

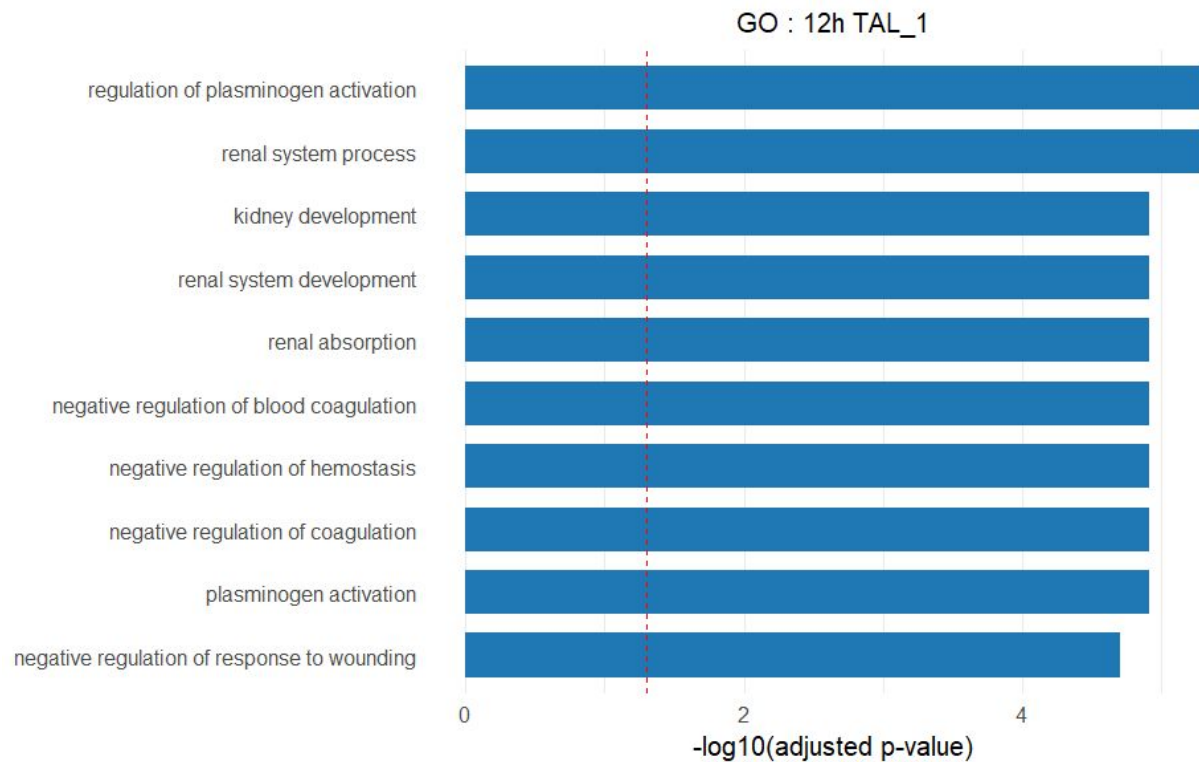


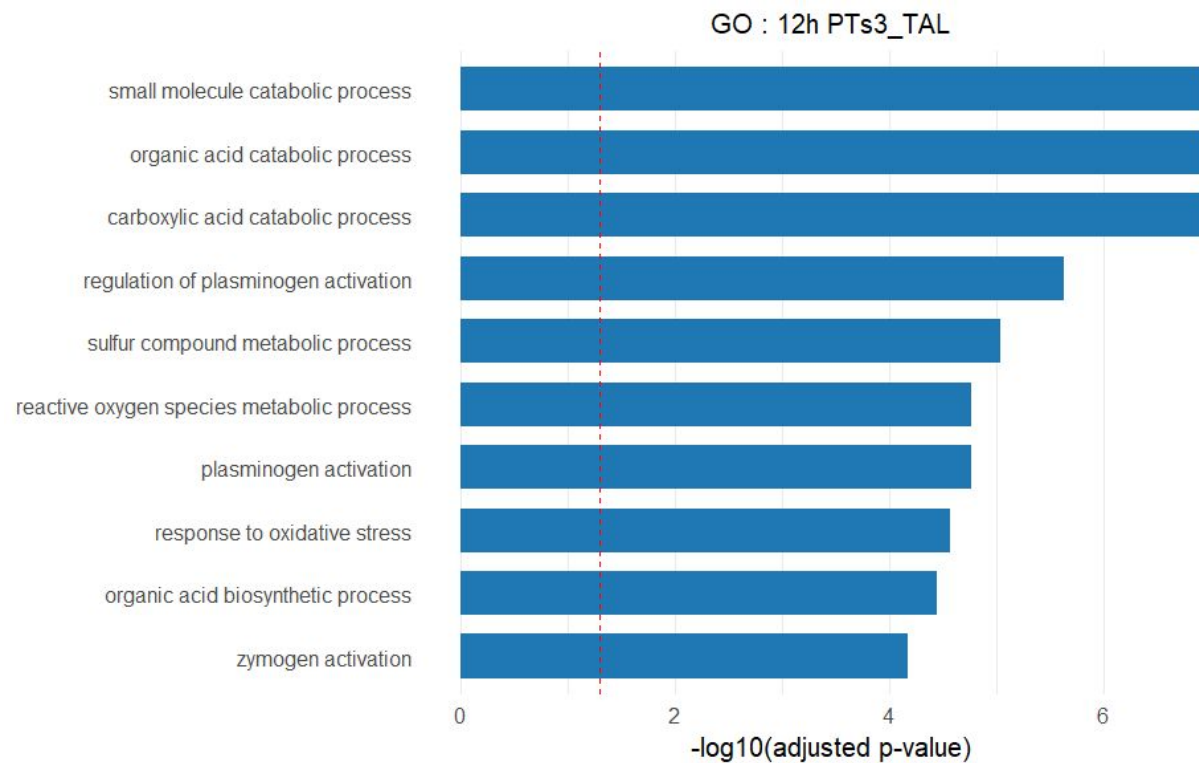
GO : 4h PTs3



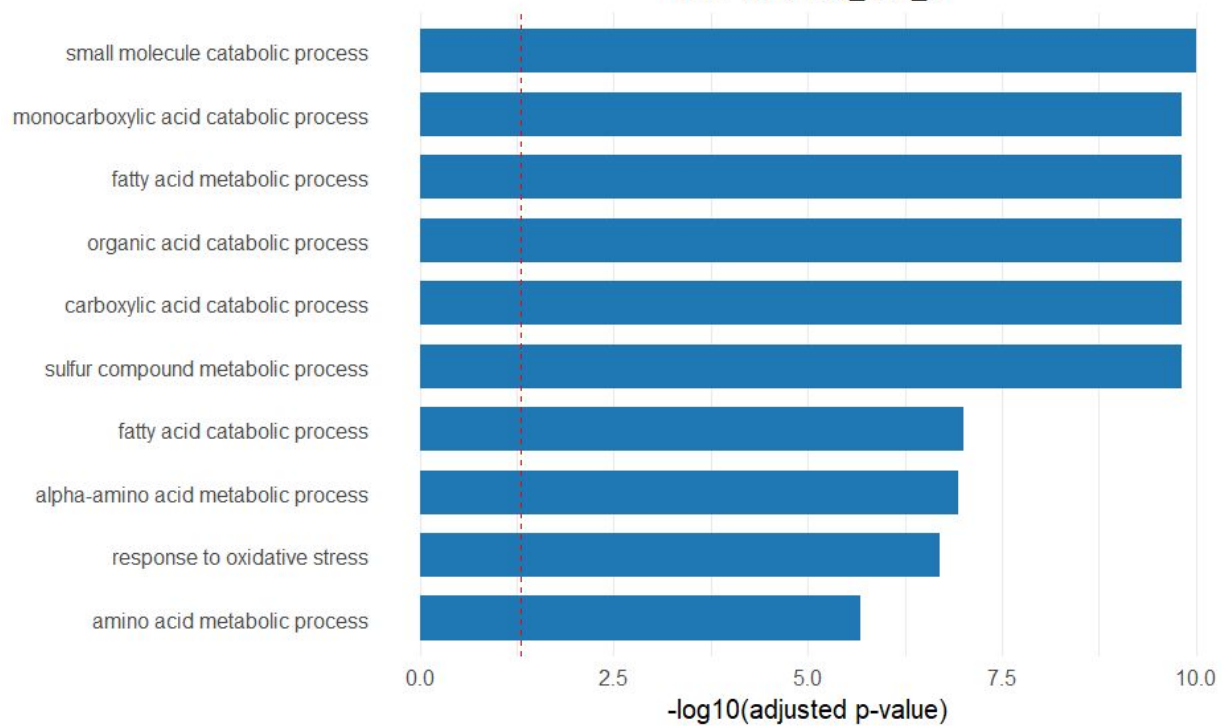








GO : 12h PTs3_TAL_1



GO : 12h PTs12_1

